

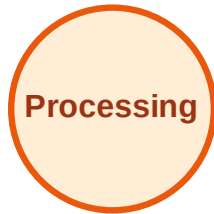
BFS Traversal

Step 1: Start at E

Legend



Complete



Processing



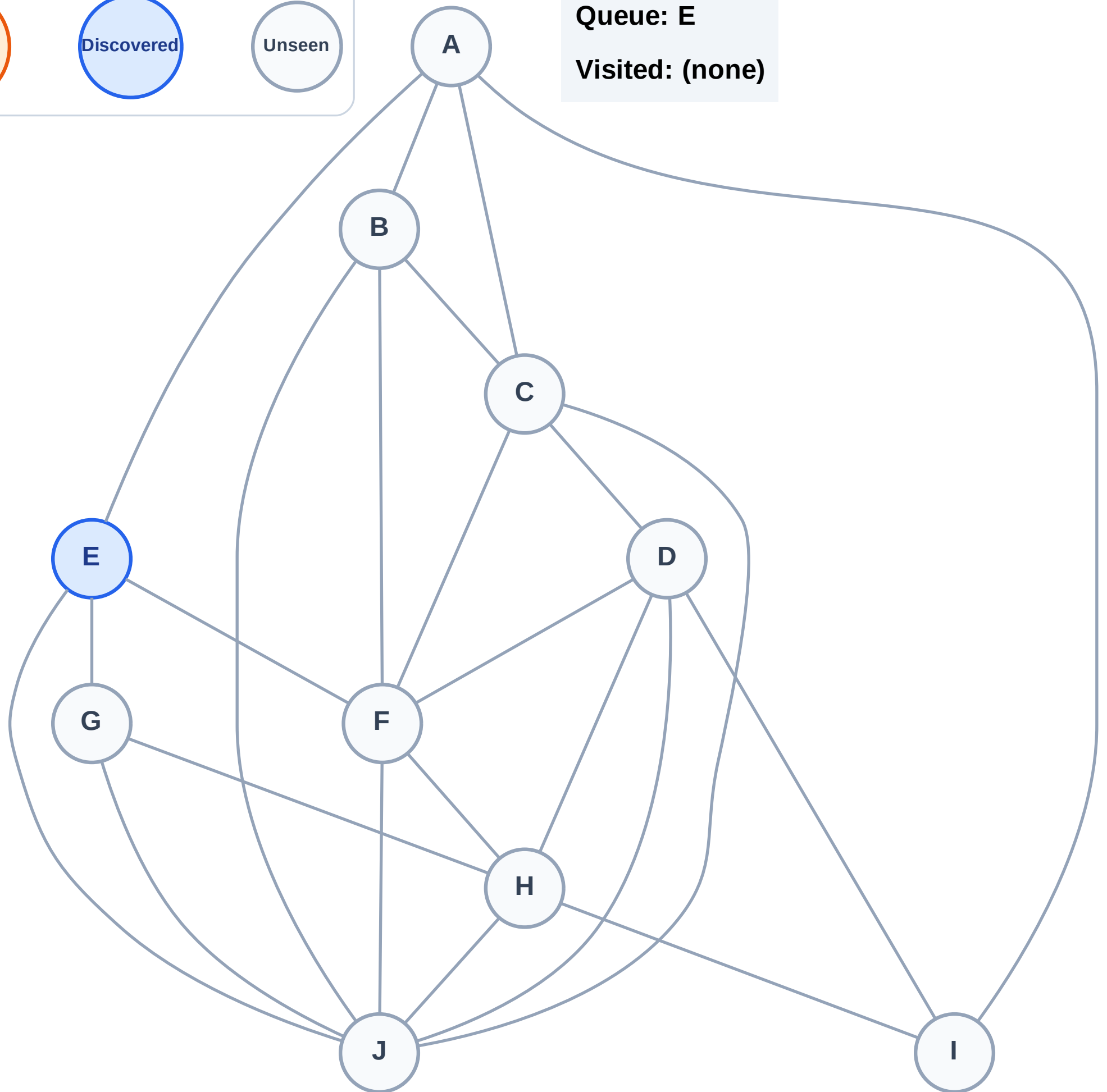
Discovered



Unseen

Queue: E

Visited: (none)



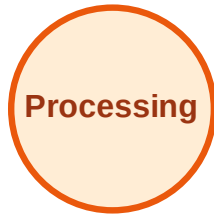
BFS Traversal

Step 2: Process node E

Legend



Complete



Processing



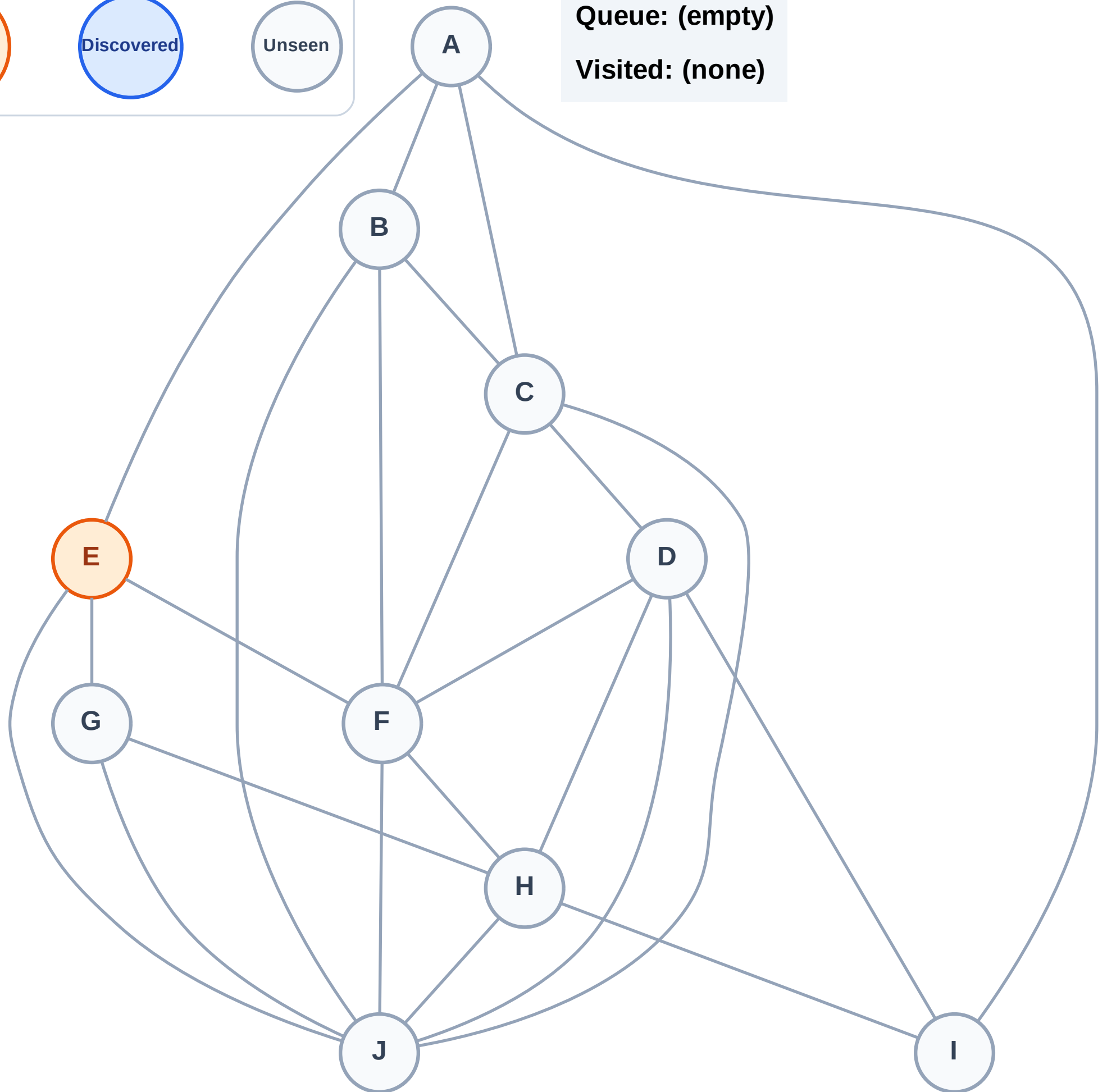
Discovered



Unseen

Queue: (empty)

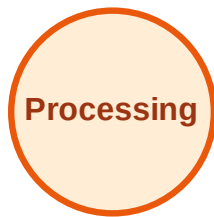
Visited: (none)



BFS Traversal

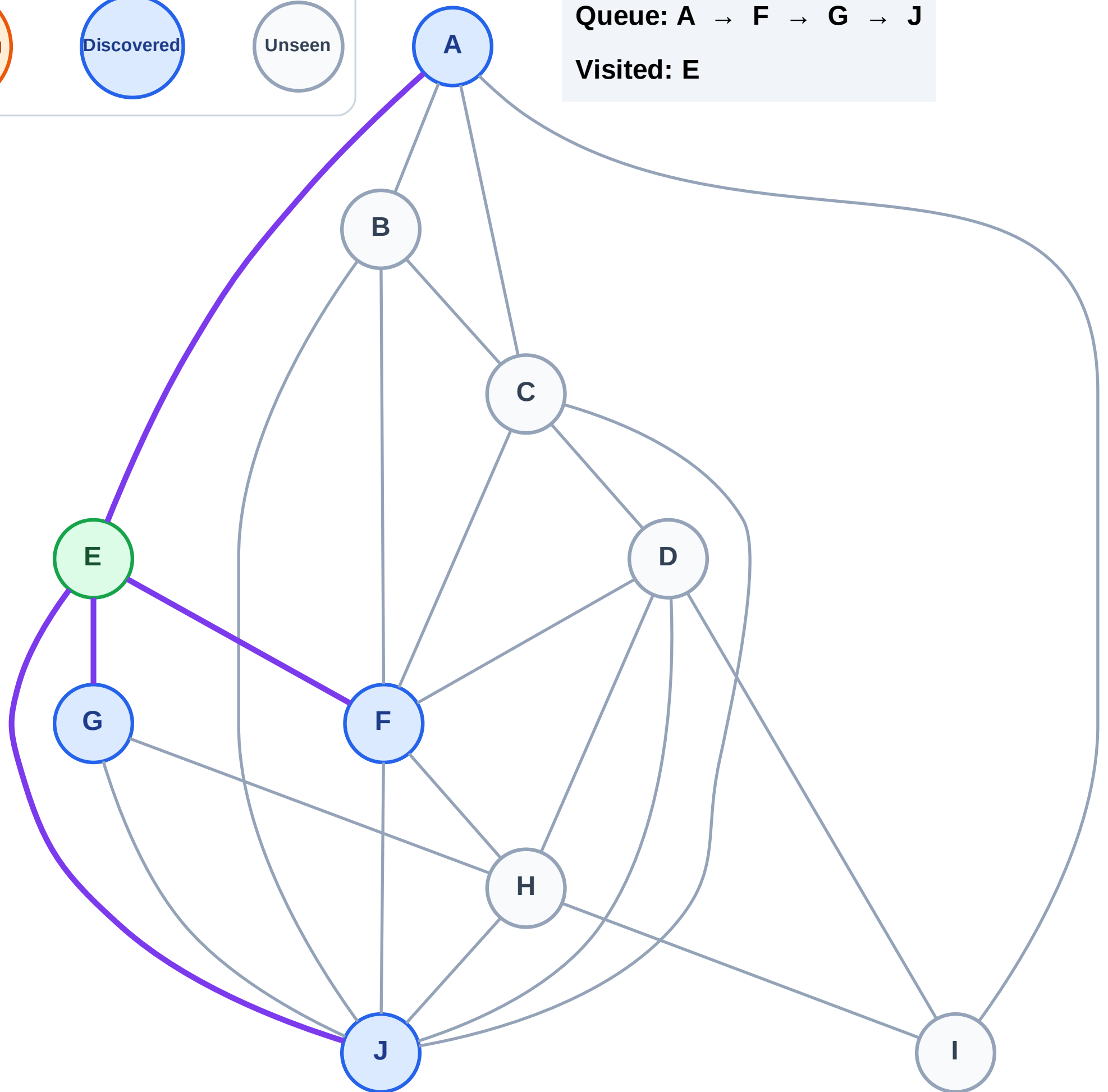
Step 3: Discover A, F, G, J from E

Legend



Queue: A → F → G → J

Visited: E



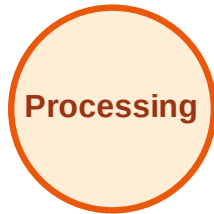
BFS Traversal

Step 4: Process node A

Legend



Complete



Processing



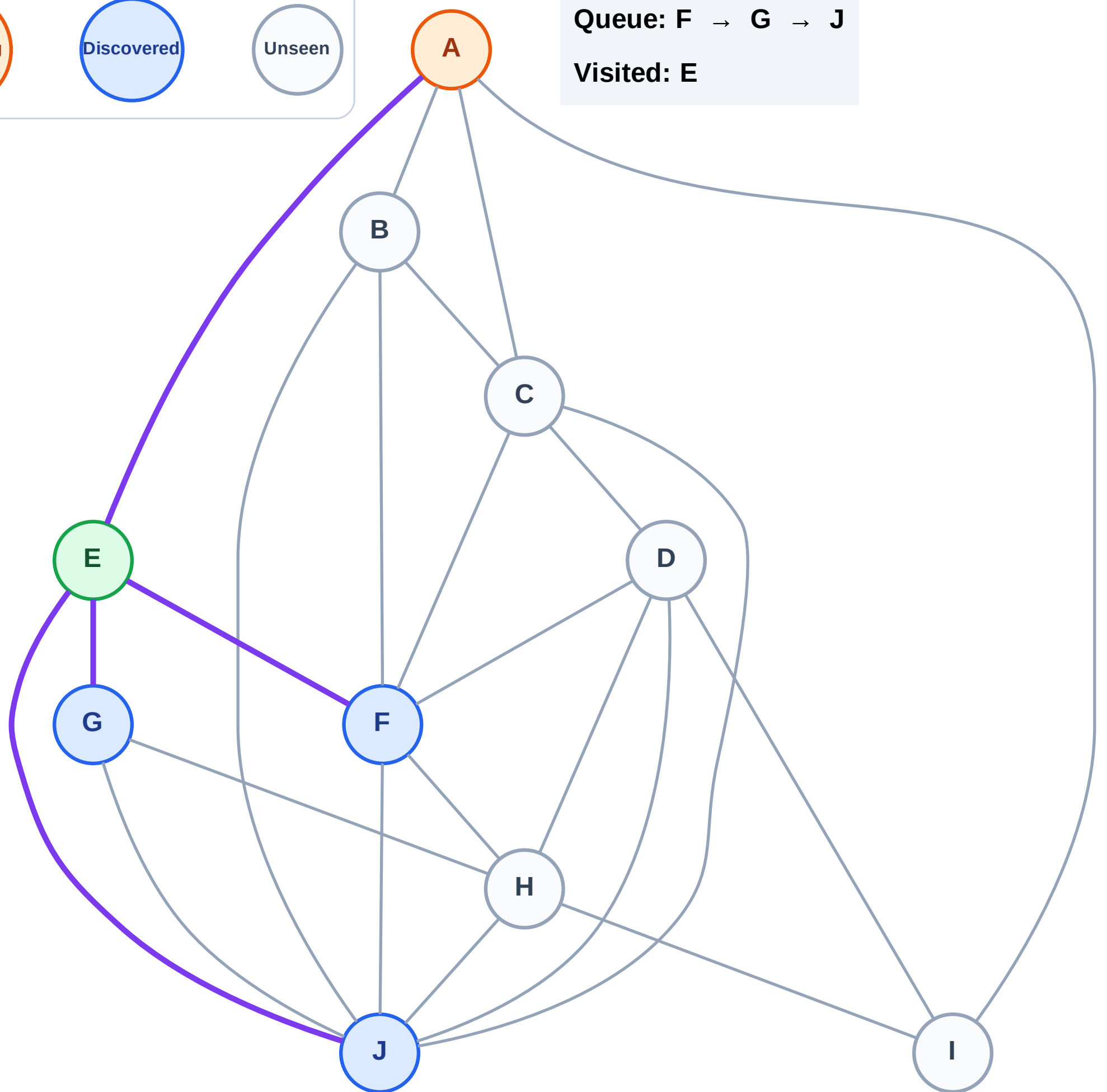
Discovered



Unseen

Queue: F → G → J

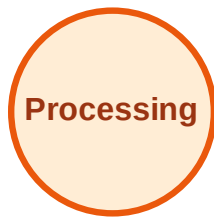
Visited: E



BFS Traversal

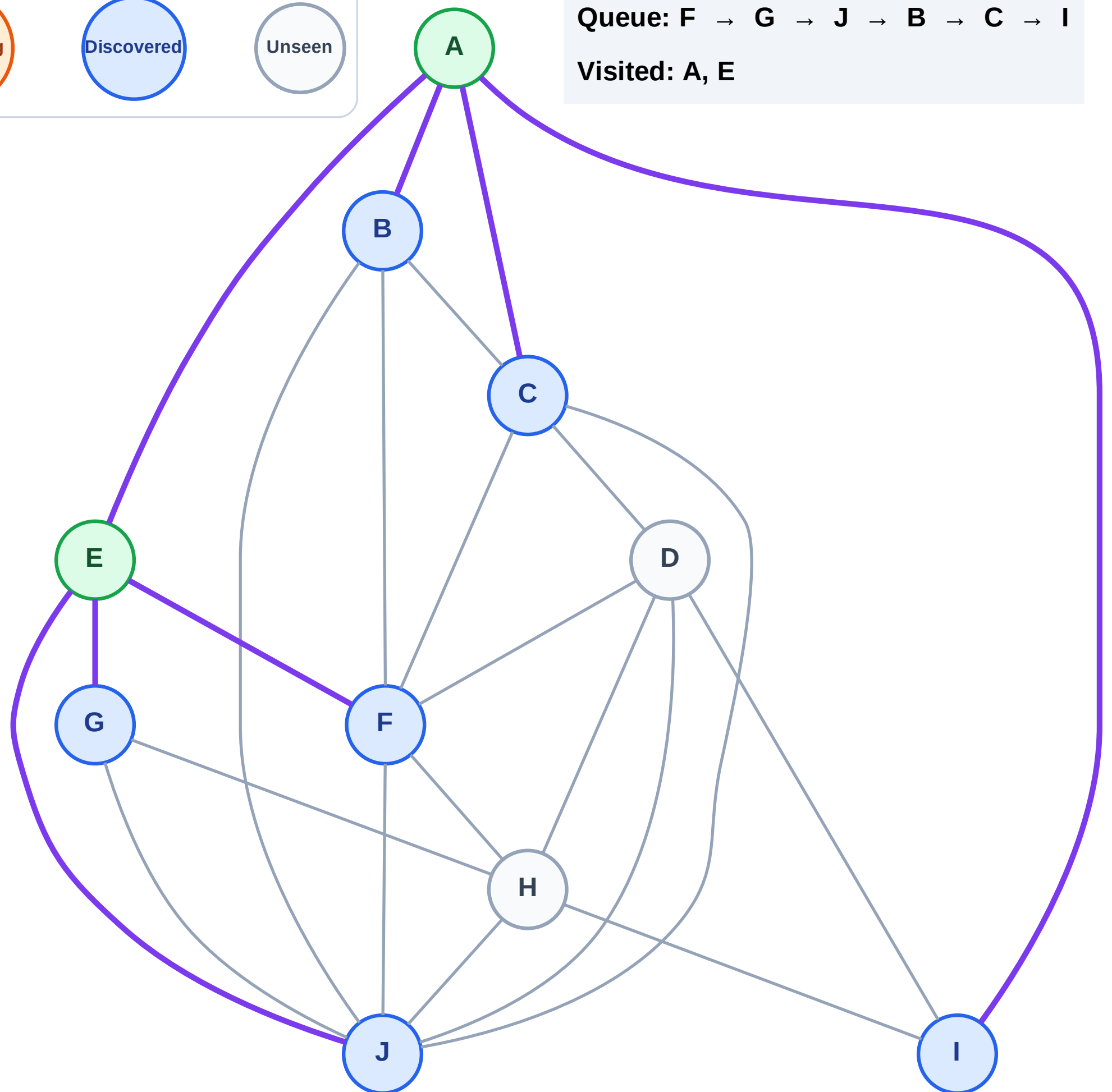
Step 5: Discover B, C, I from A

Legend



Queue: F → G → J → B → C → I

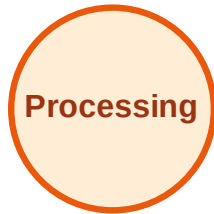
Visited: A, E



BFS Traversal

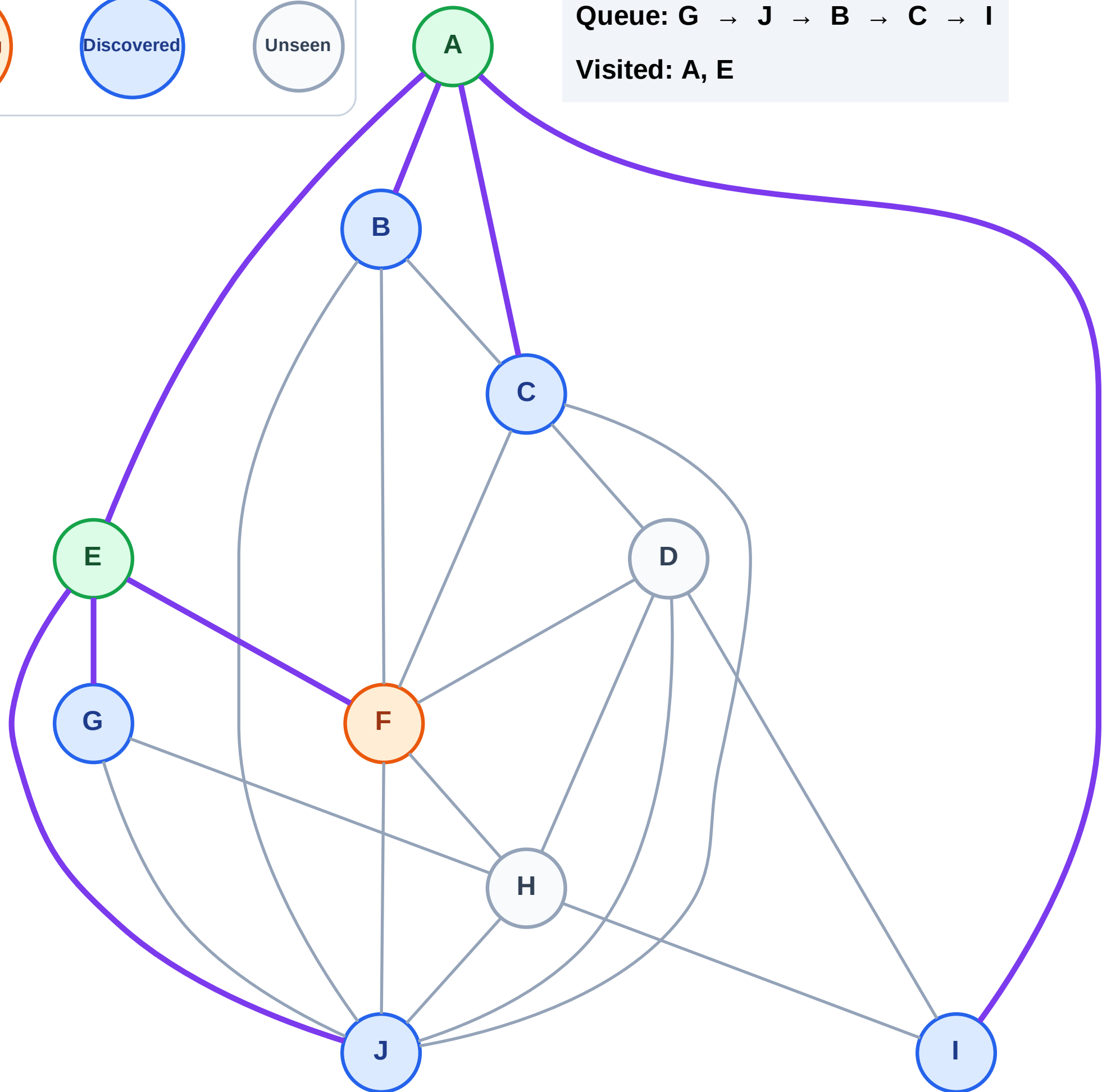
Step 6: Process node F

Legend



Queue: G → J → B → C → I

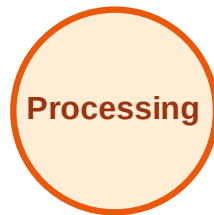
Visited: A, E



BFS Traversal

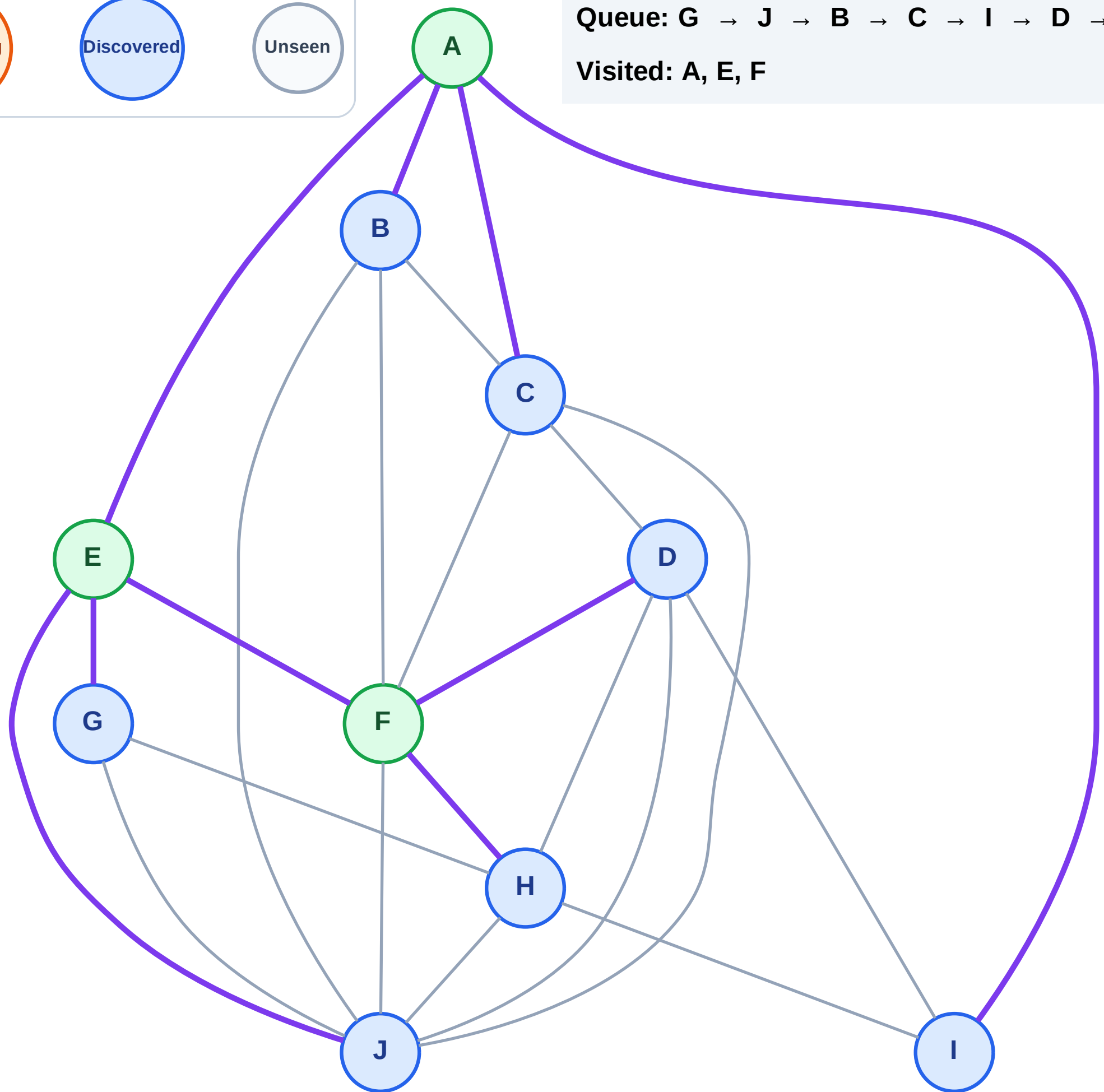
Step 7: Discover D, H from F

Legend



Queue: G → J → B → C → I → D → H

Visited: A, E, F



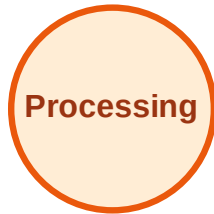
BFS Traversal

Step 8: Process node G

Legend



Complete



Processing



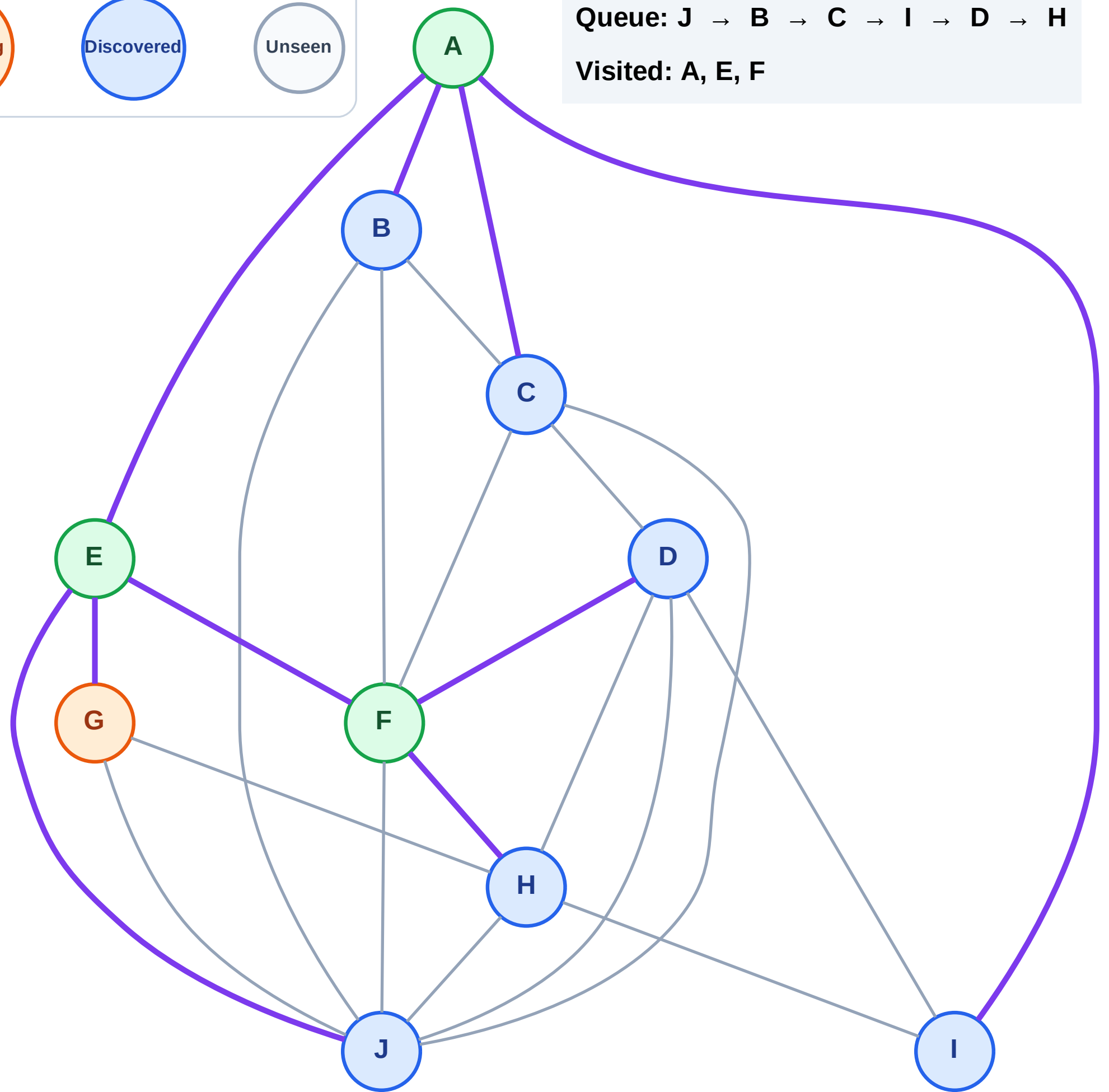
Discovered



Unseen

Queue: J → B → C → I → D → H

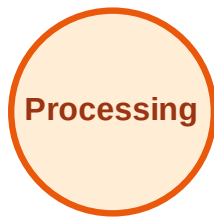
Visited: A, E, F



BFS Traversal

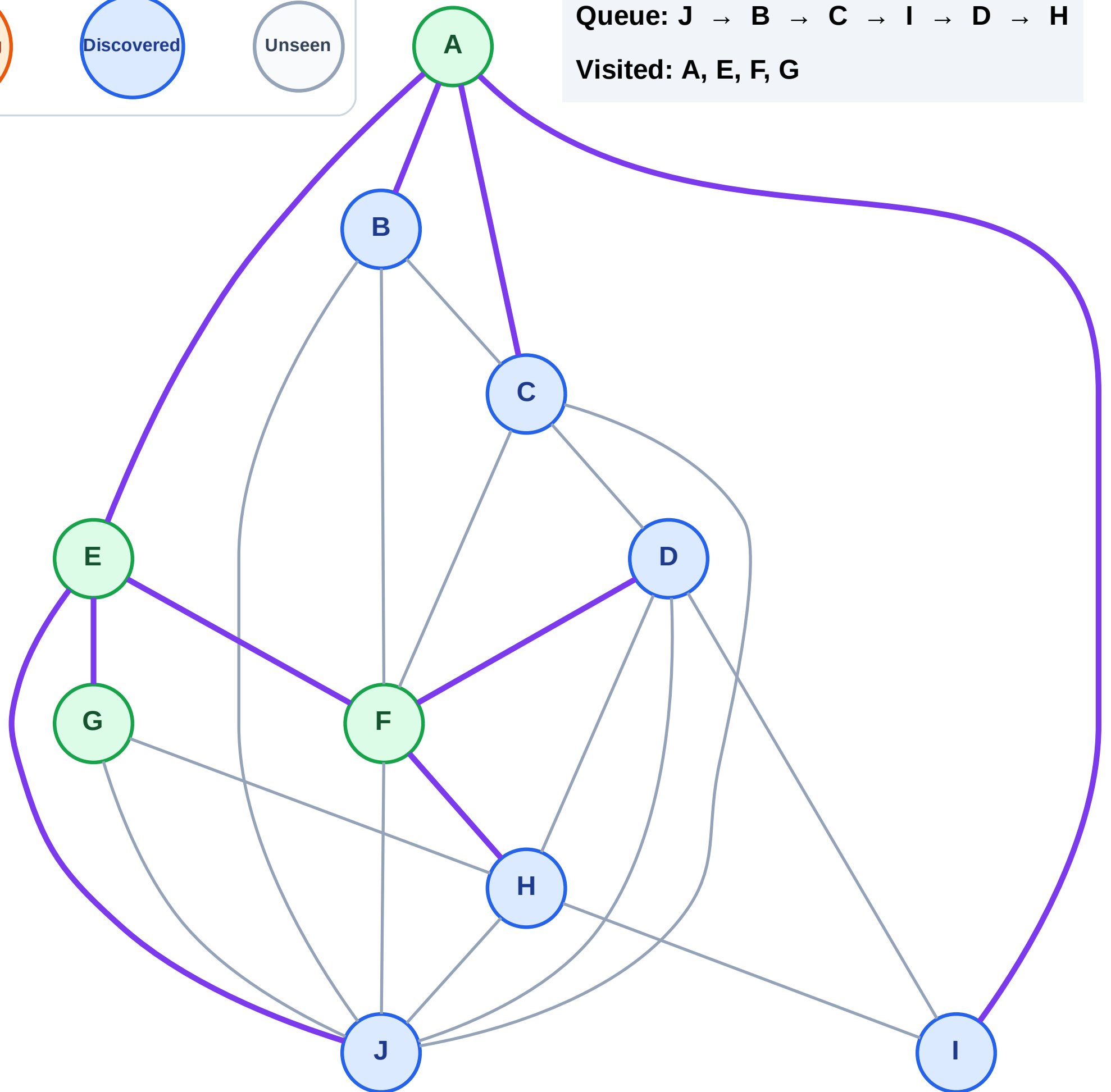
Step 9: No new neighbors from G

Legend



Queue: J → B → C → I → D → H

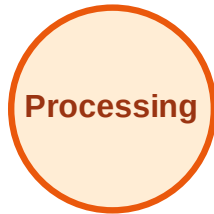
Visited: A, E, F, G



BFS Traversal

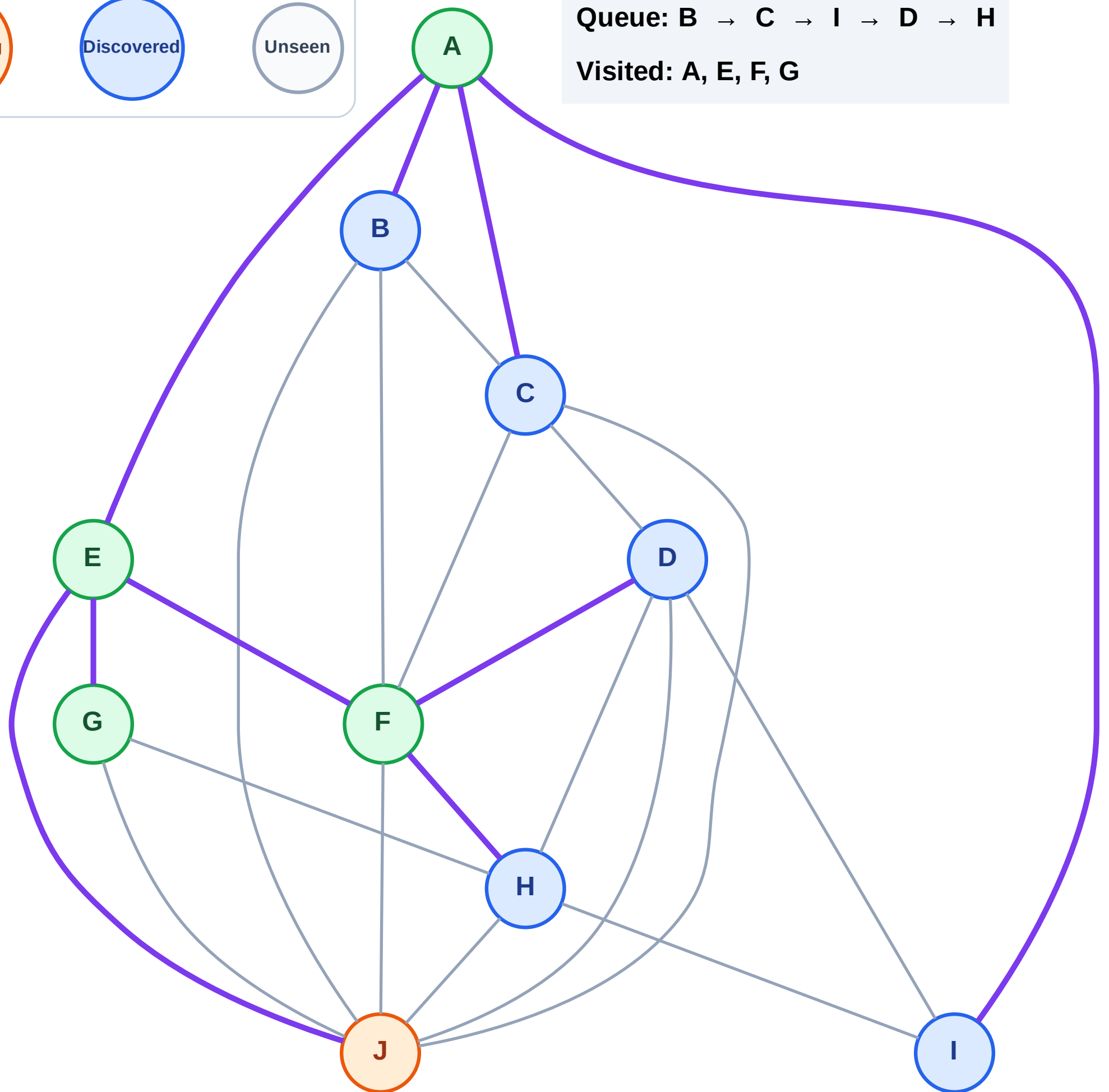
Step 10: Process node J

Legend



Queue: B → C → I → D → H

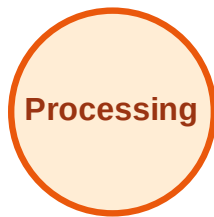
Visited: A, E, F, G



BFS Traversal

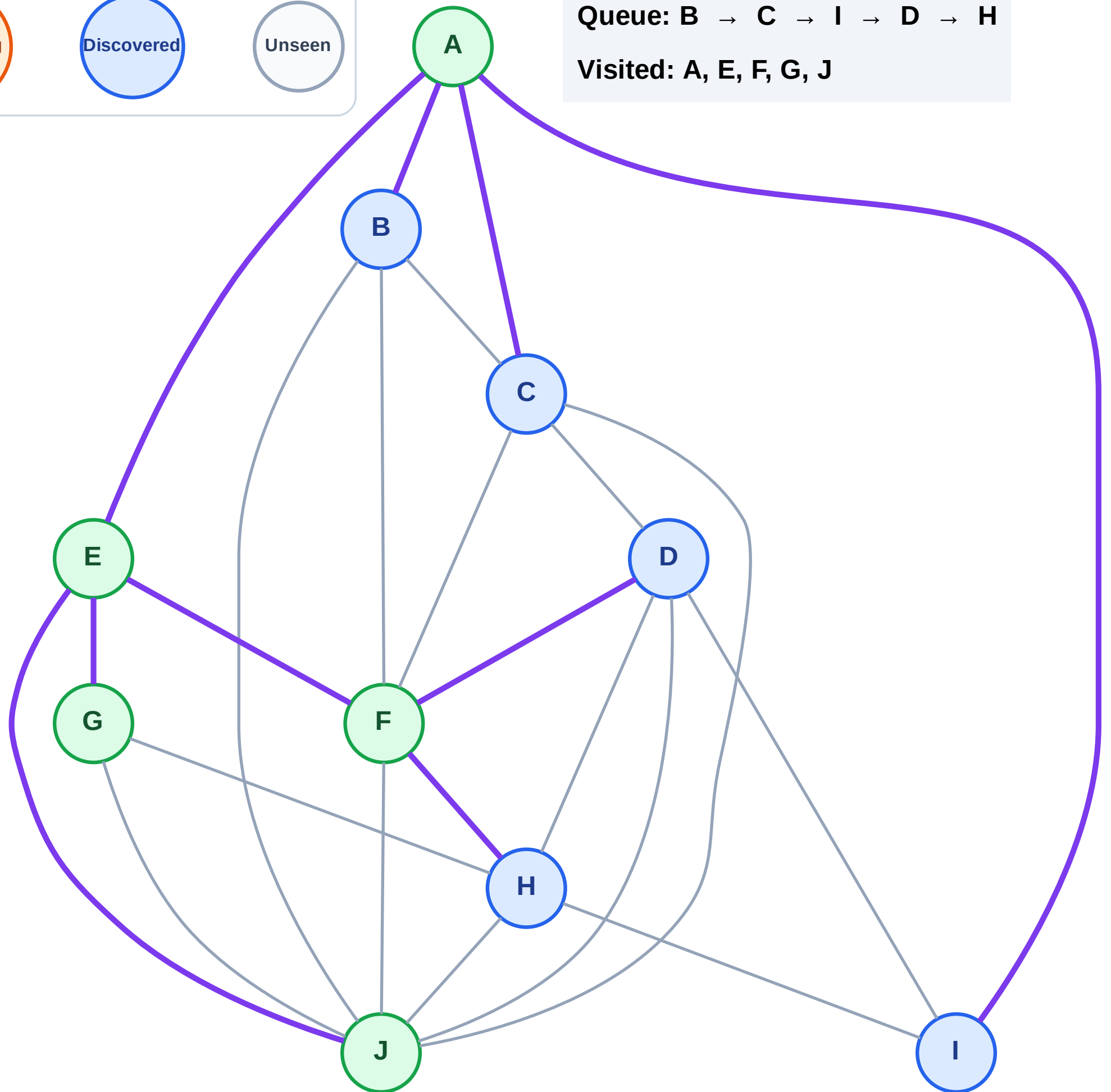
Step 11: No new neighbors from J

Legend



Queue: B → C → I → D → H

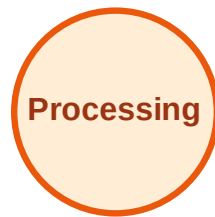
Visited: A, E, F, G, J



BFS Traversal

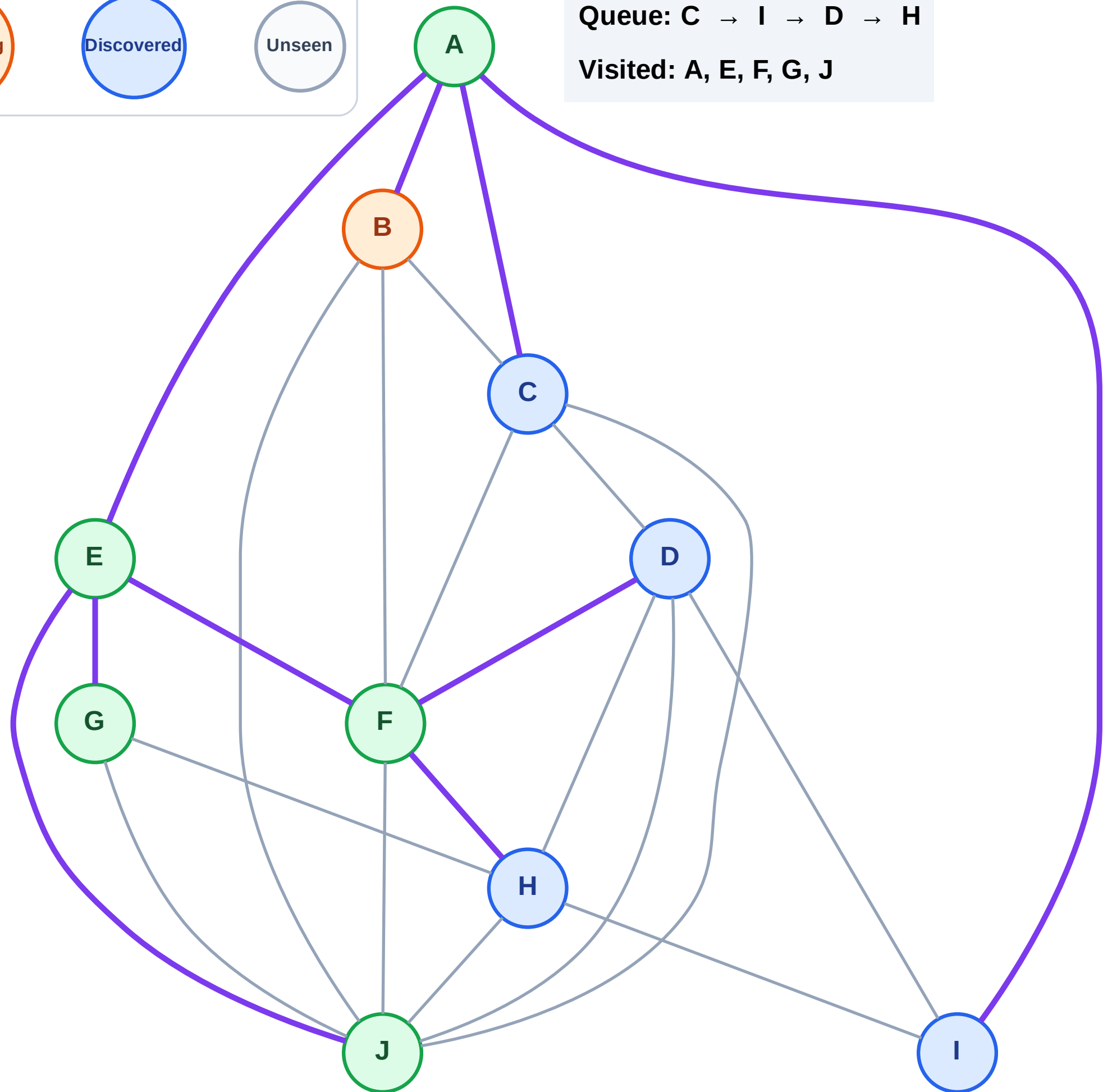
Step 12: Process node B

Legend



Queue: C → I → D → H

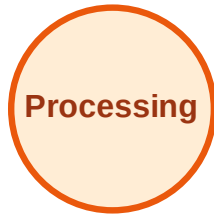
Visited: A, E, F, G, J



BFS Traversal

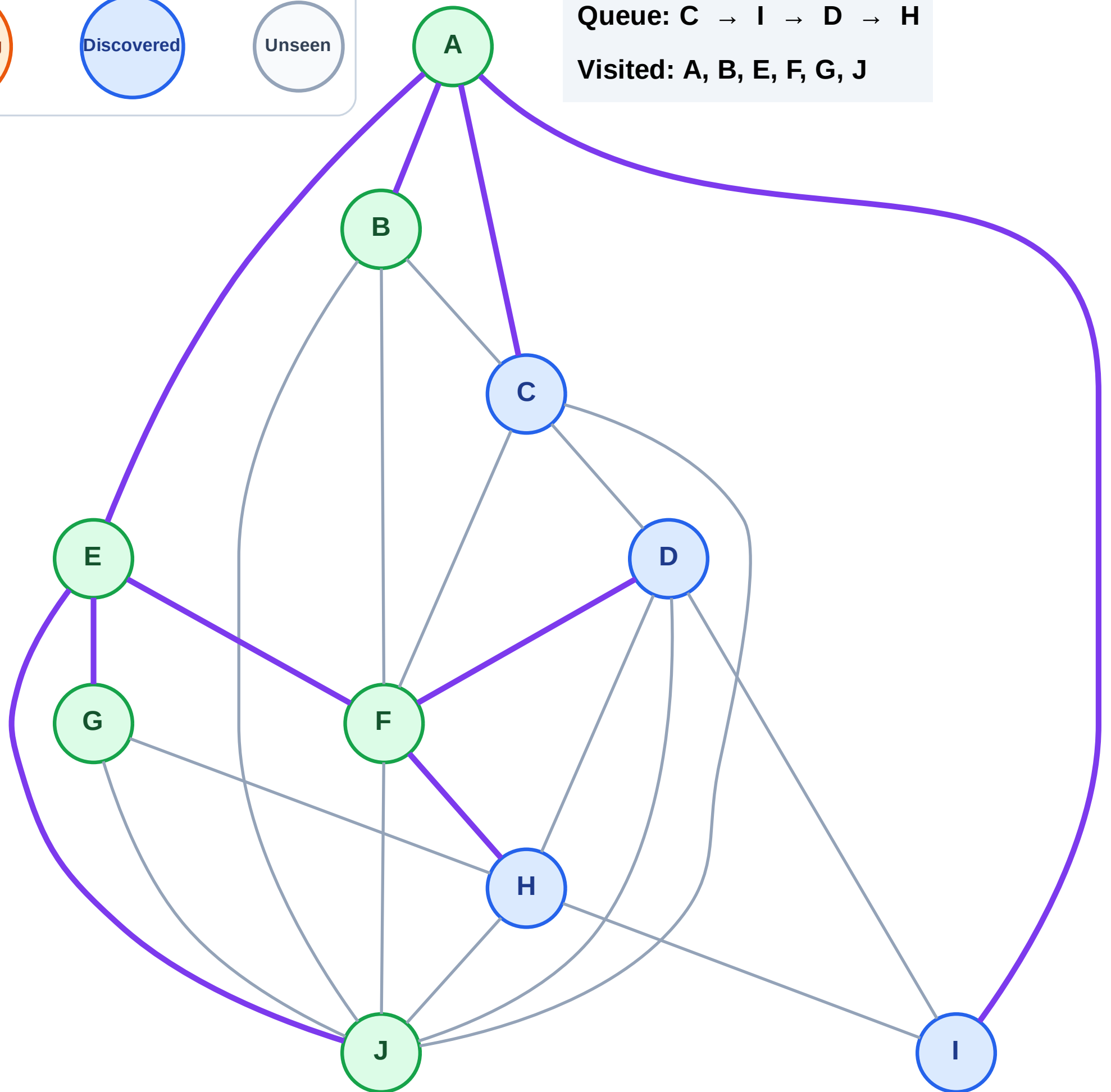
Step 13: No new neighbors from B

Legend



Queue: C → I → D → H

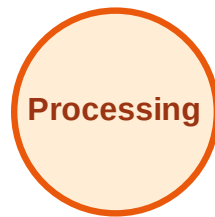
Visited: A, B, E, F, G, J



BFS Traversal

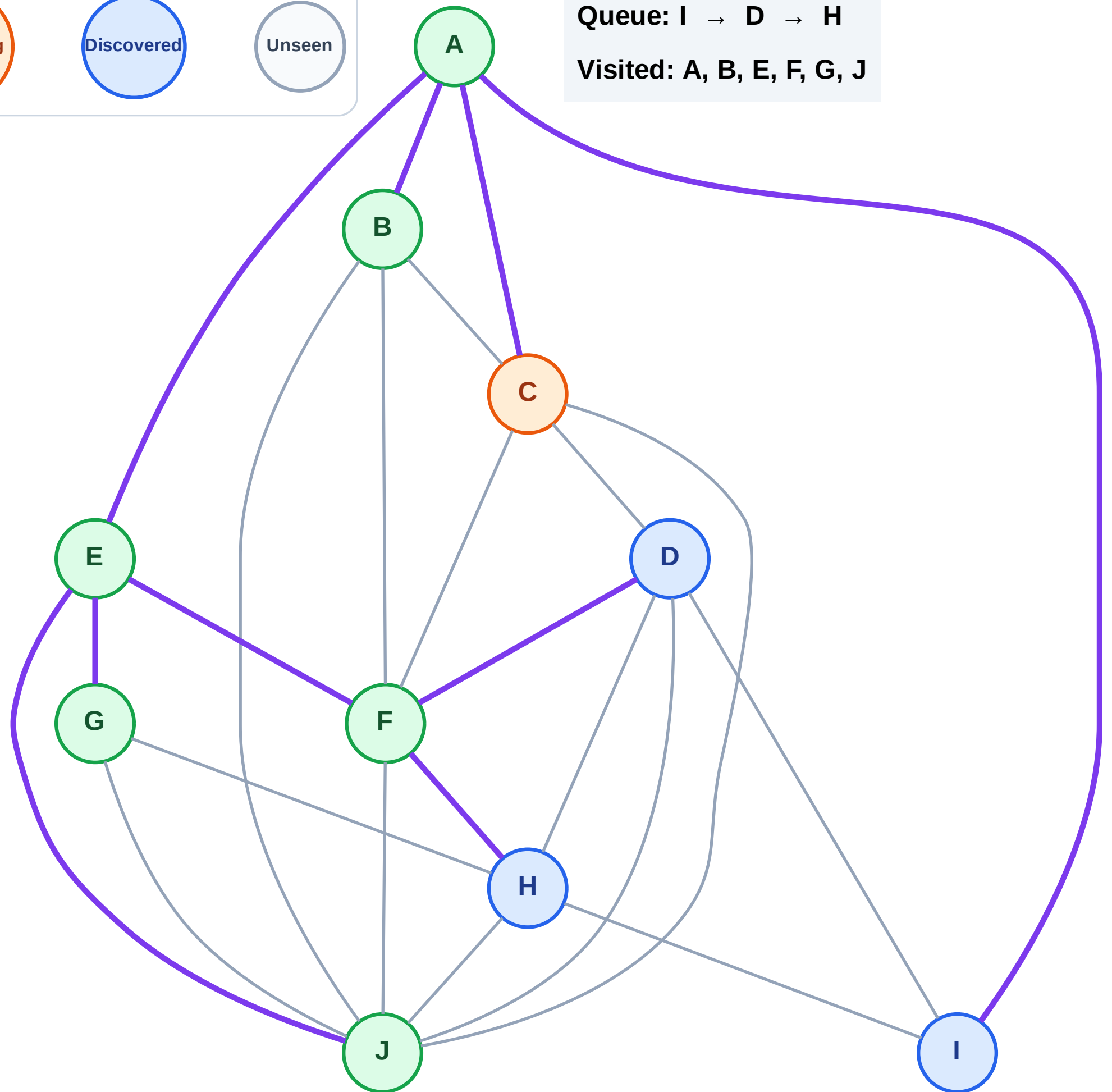
Step 14: Process node C

Legend



Queue: I → D → H

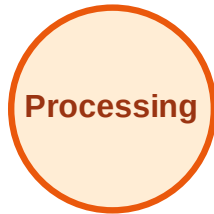
Visited: A, B, E, F, G, J



BFS Traversal

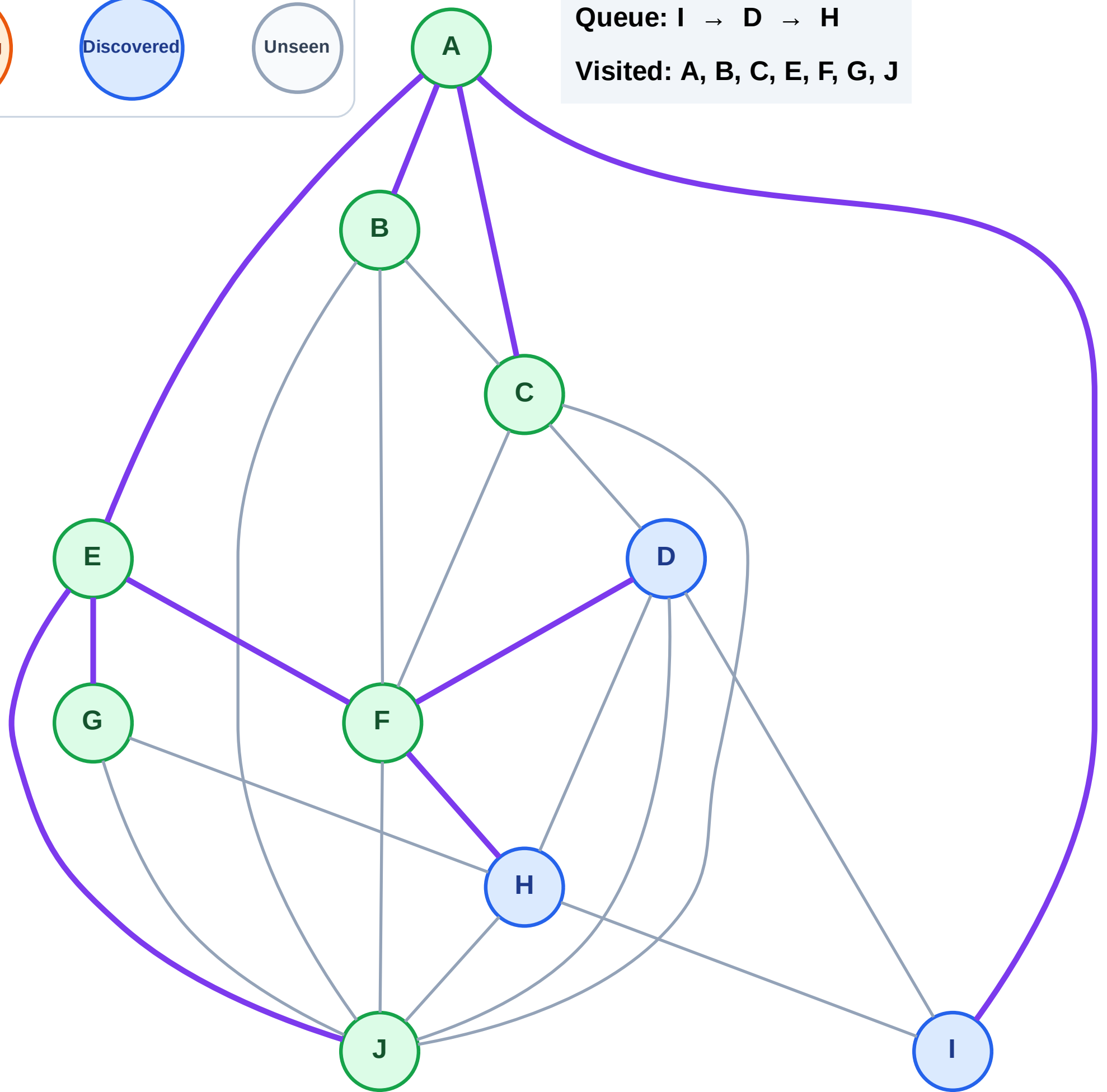
Step 15: No new neighbors from C

Legend



Queue: I → D → H

Visited: A, B, C, E, F, G, J



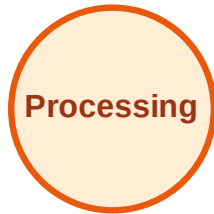
BFS Traversal

Step 16: Process node I

Legend



Complete



Processing



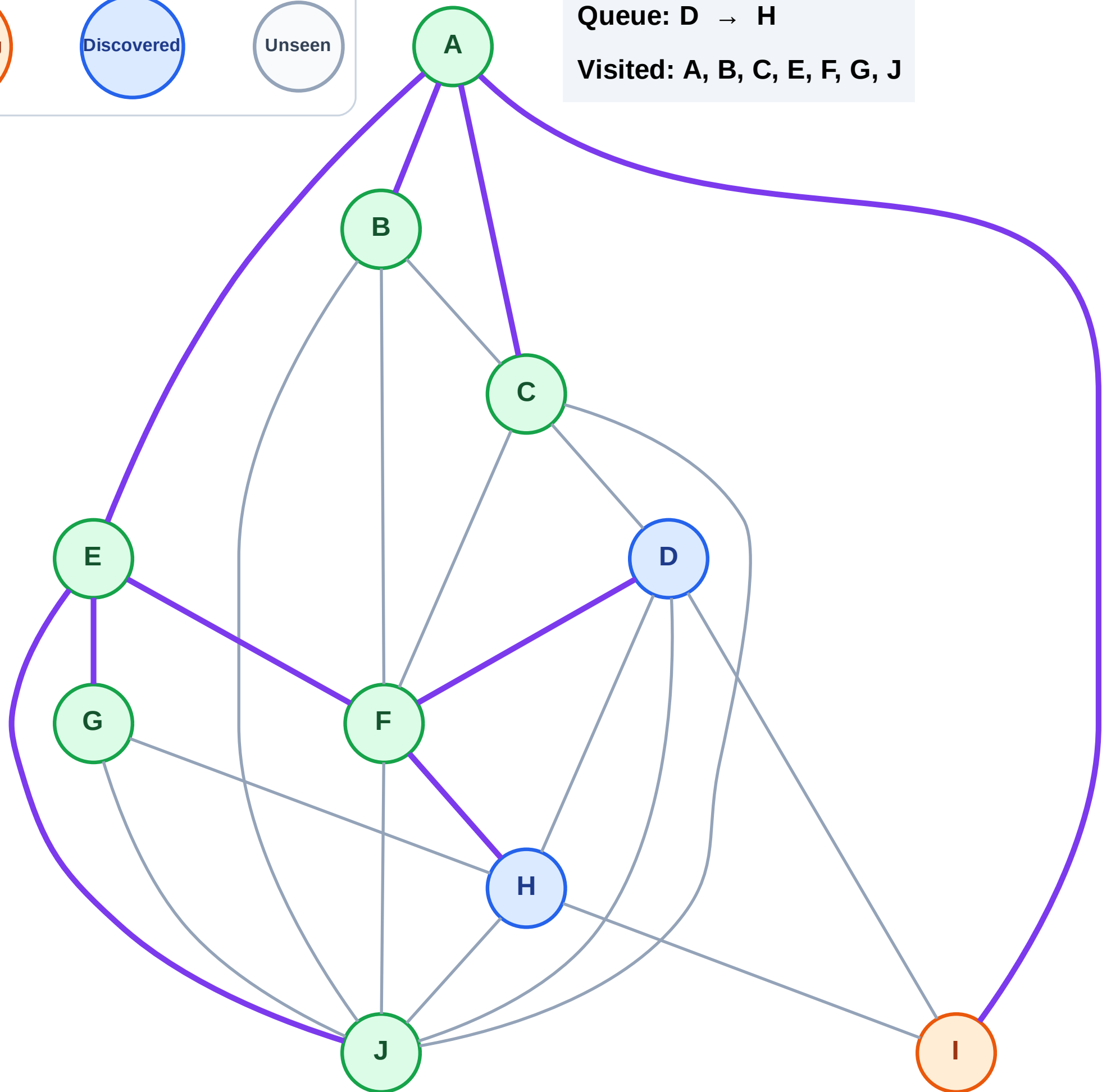
Discovered



Unseen

Queue: D → H

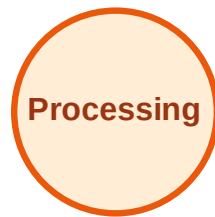
Visited: A, B, C, E, F, G, J



BFS Traversal

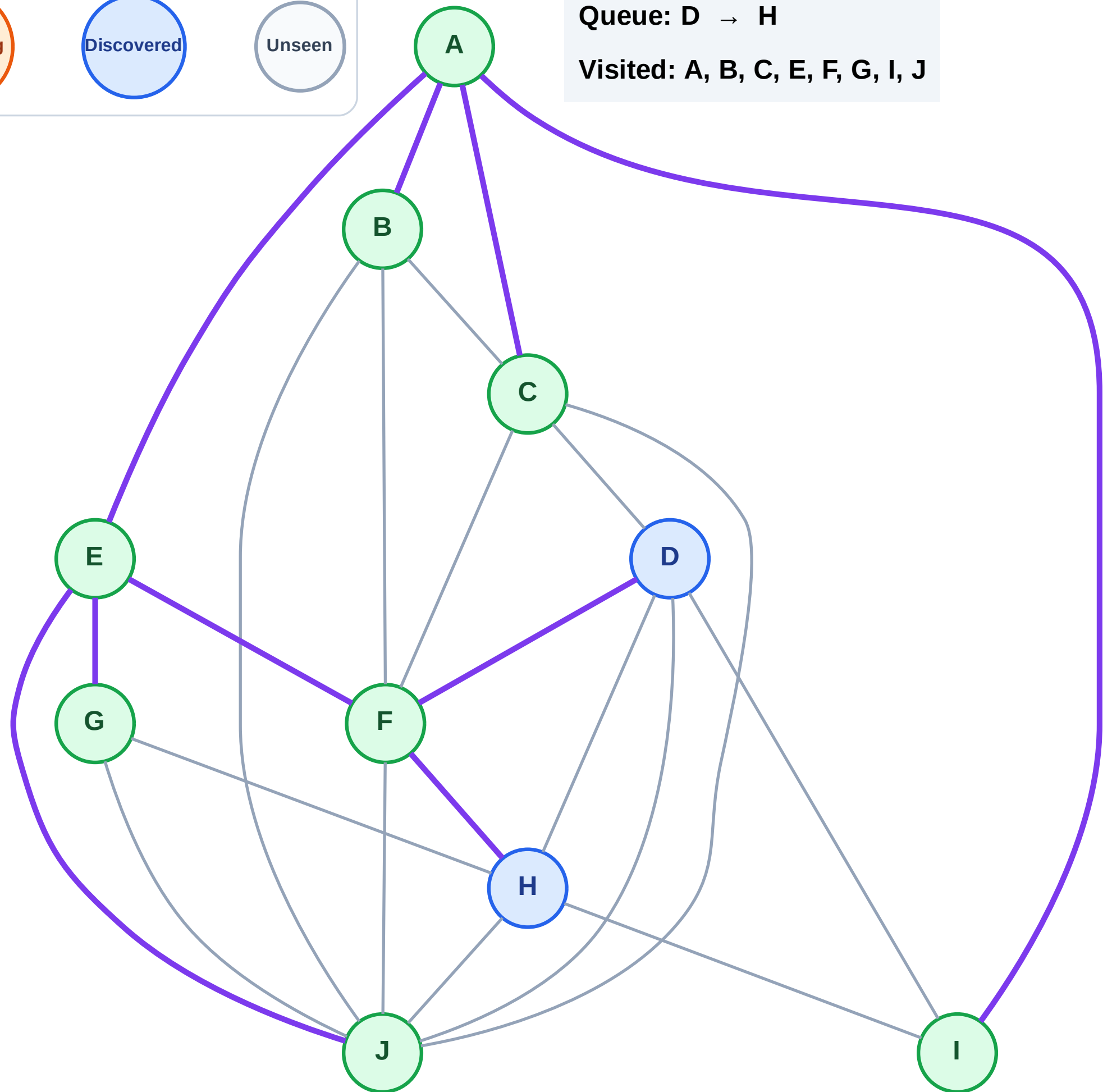
Step 17: No new neighbors from I

Legend



Queue: D → H

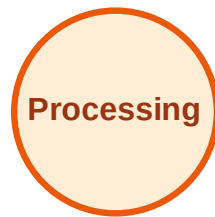
Visited: A, B, C, E, F, G, I, J



BFS Traversal

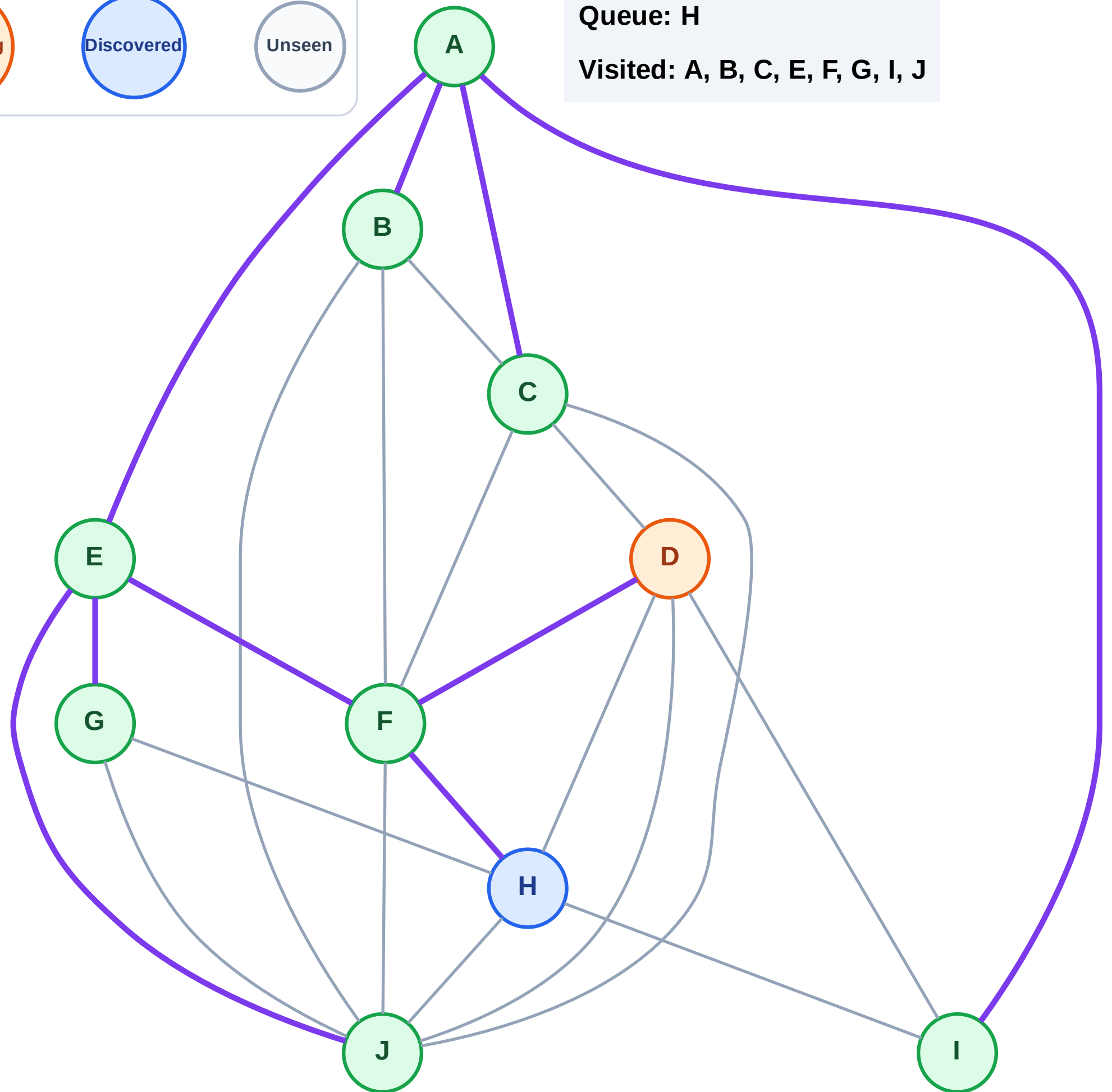
Step 18: Process node D

Legend



Queue: H

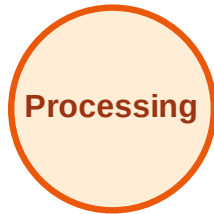
Visited: A, B, C, E, F, G, I, J



BFS Traversal

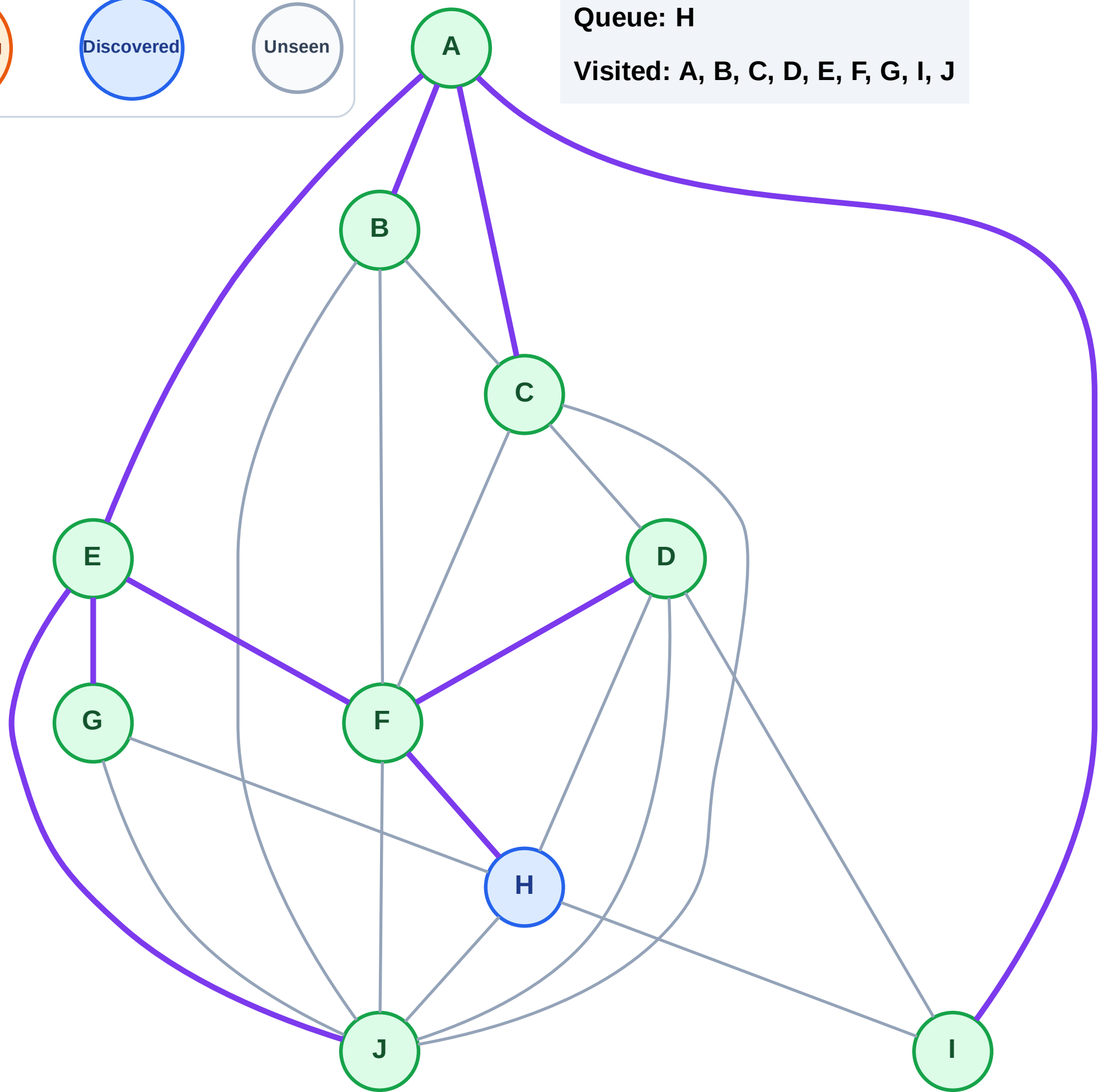
Step 19: No new neighbors from D

Legend



Queue: H

Visited: A, B, C, D, E, F, G, I, J



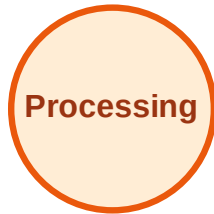
BFS Traversal

Step 20: Process node H

Legend



Complete



Processing



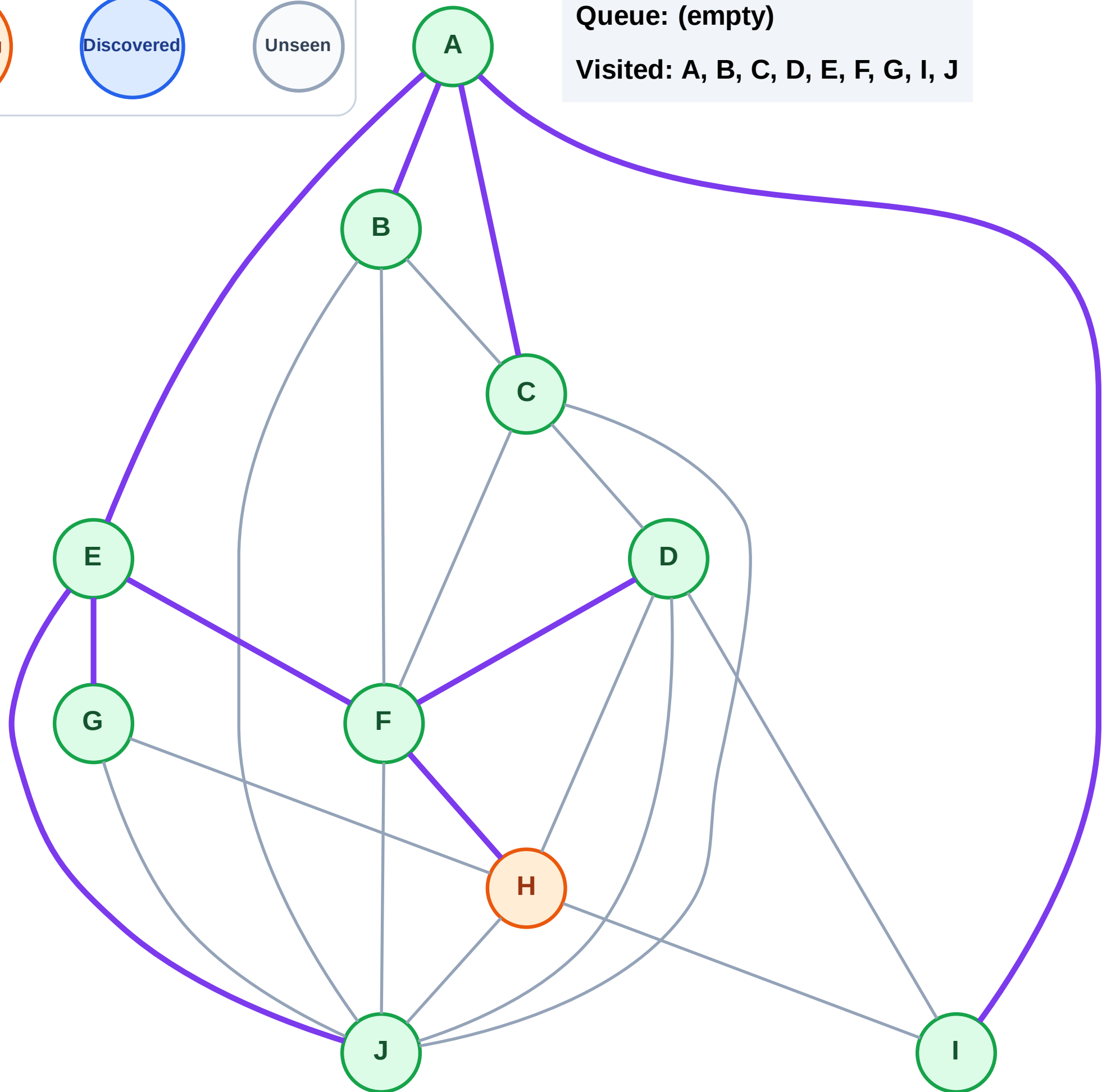
Discovered



Unseen

Queue: (empty)

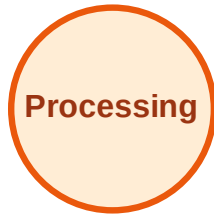
Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

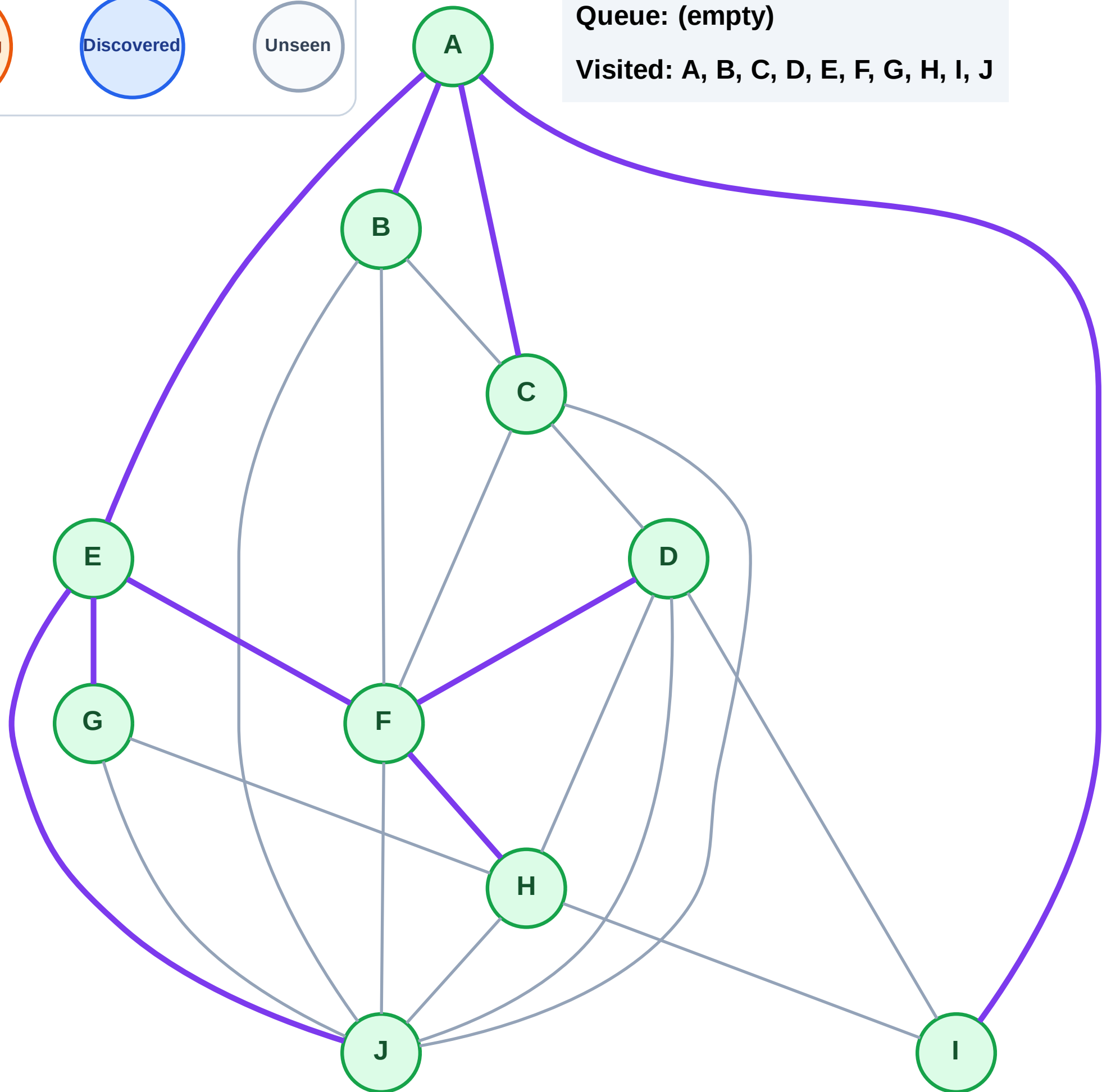
Step 21: No new neighbors from H

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



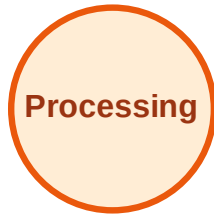
DFS Traversal

Step 1: Start at E

Legend



Complete



Processing



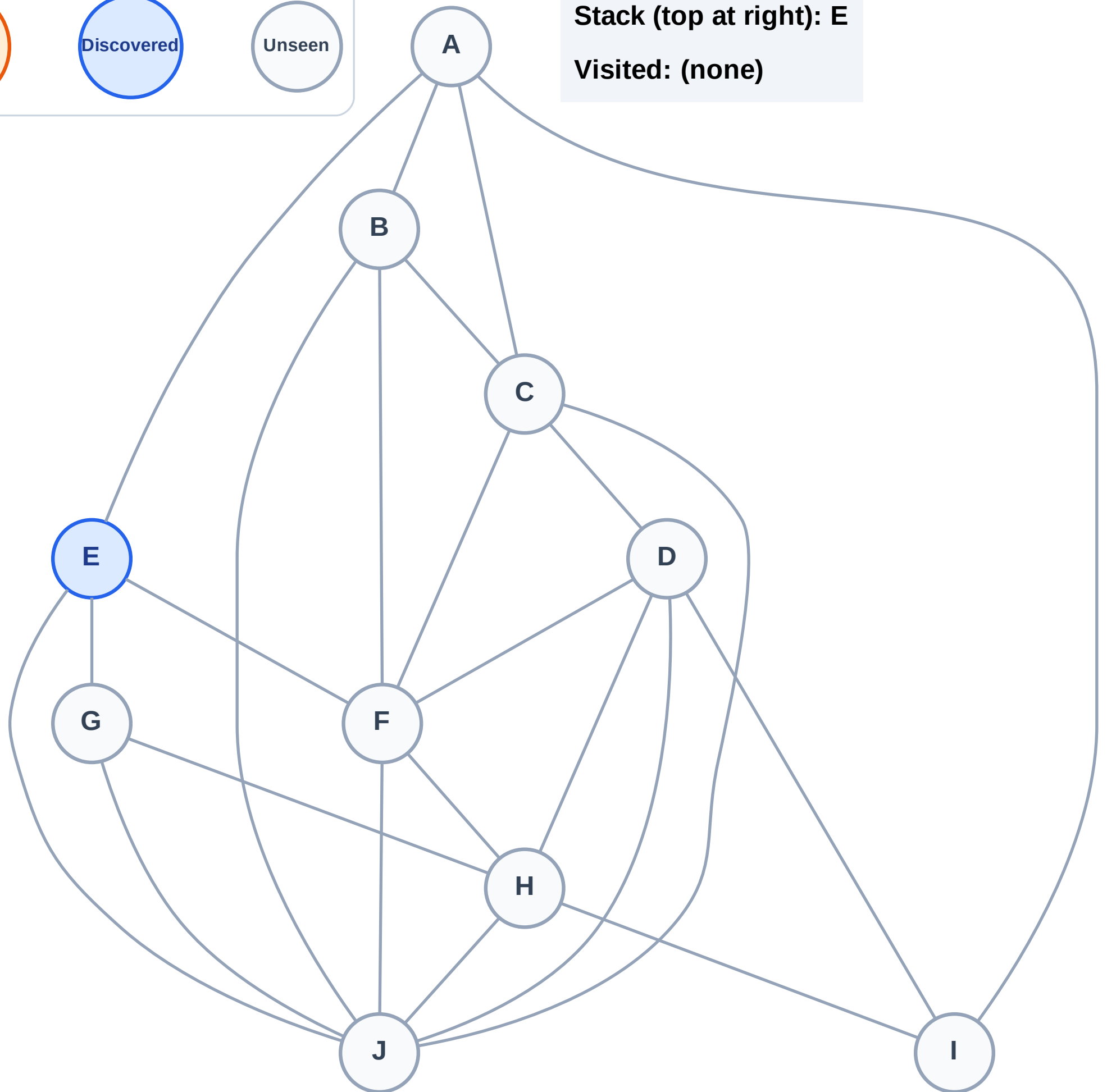
Discovered



Unseen

Stack (top at right): E

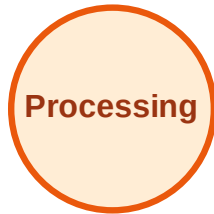
Visited: (none)



DFS Traversal

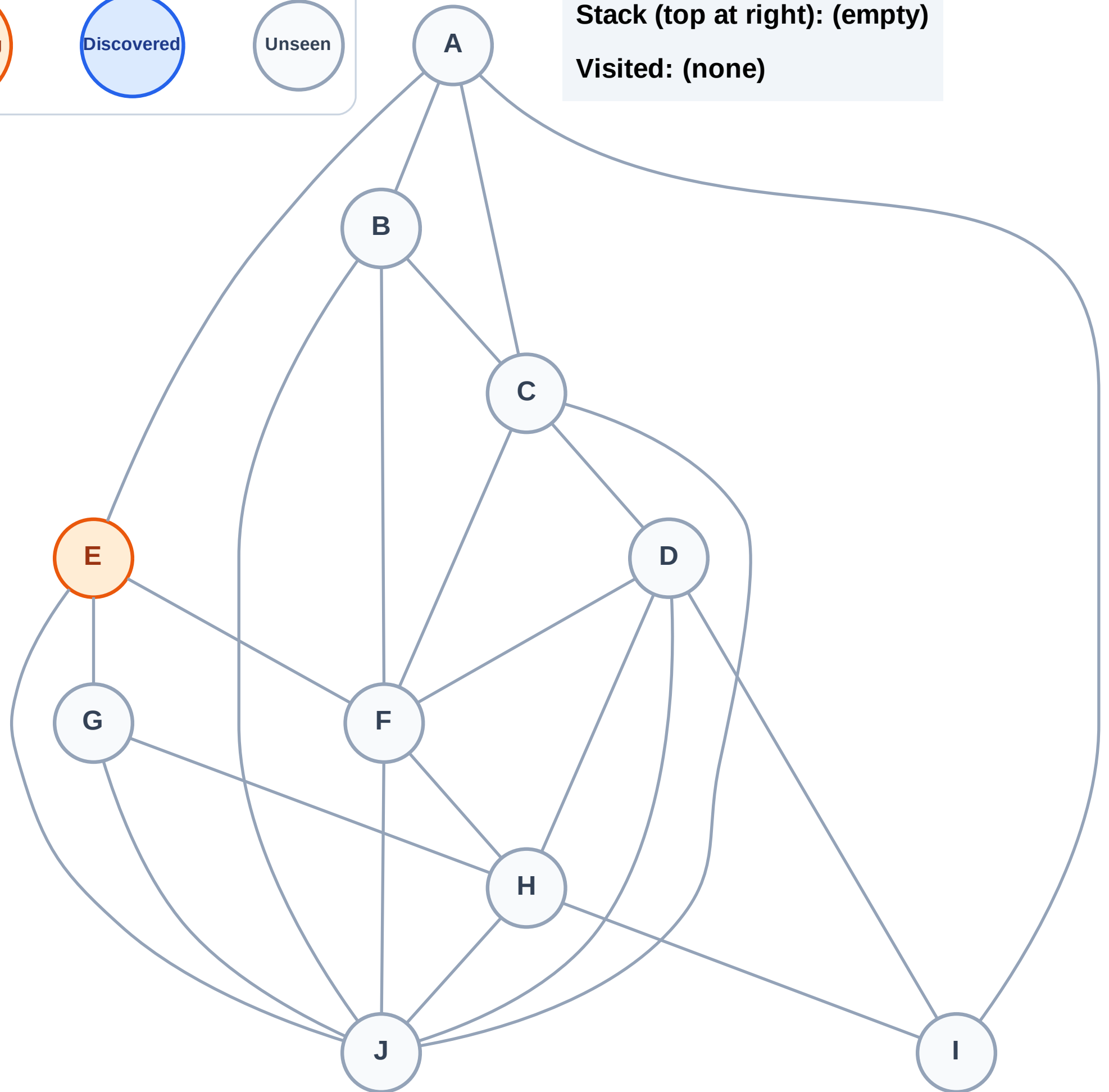
Step 2: Process node E

Legend



Stack (top at right): (empty)

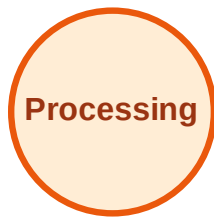
Visited: (none)



DFS Traversal

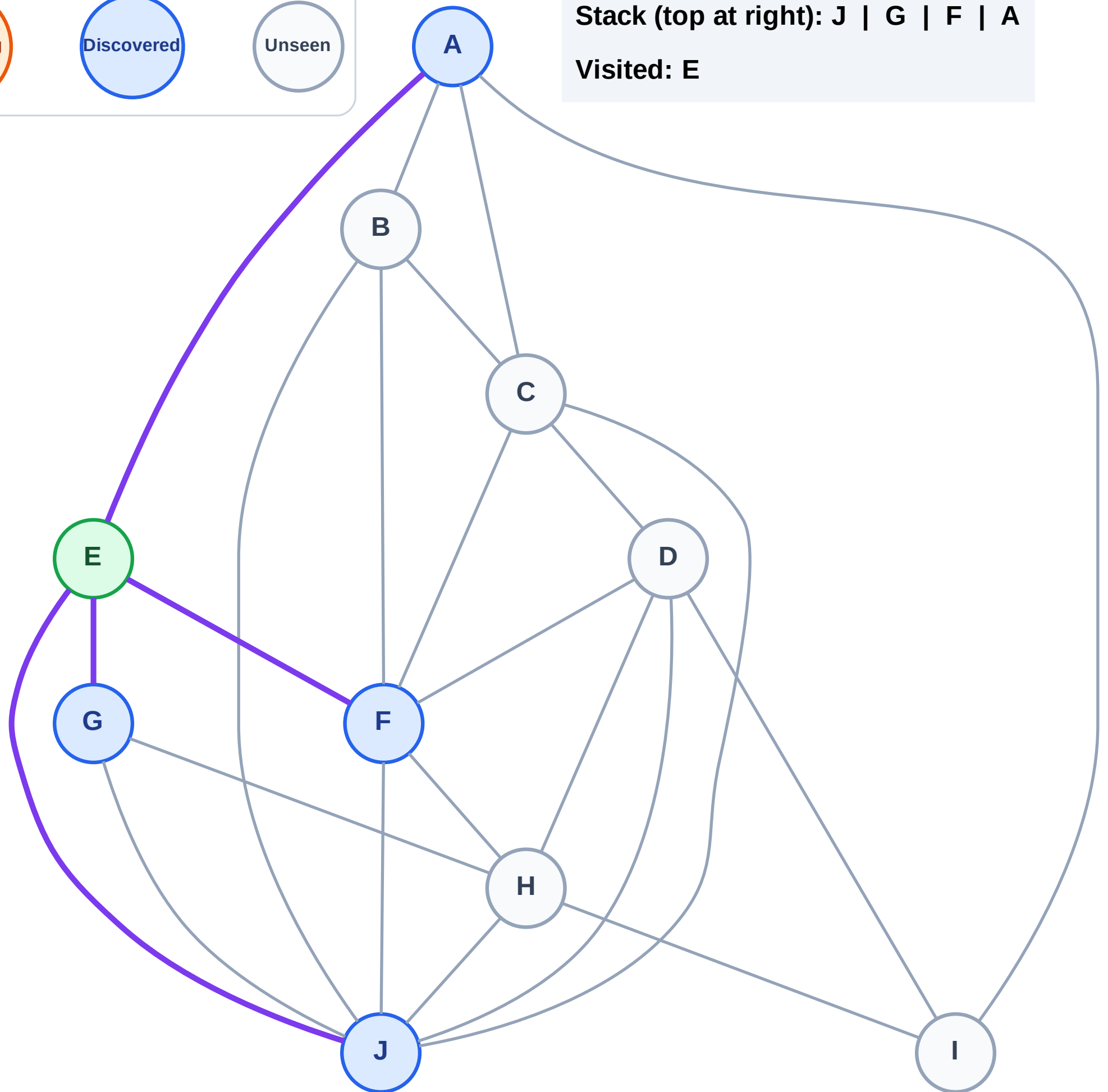
Step 3: Discover J, G, F, A from E

Legend



Stack (top at right): J | G | F | A

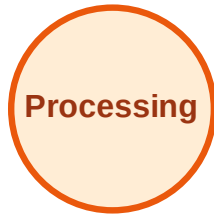
Visited: E



DFS Traversal

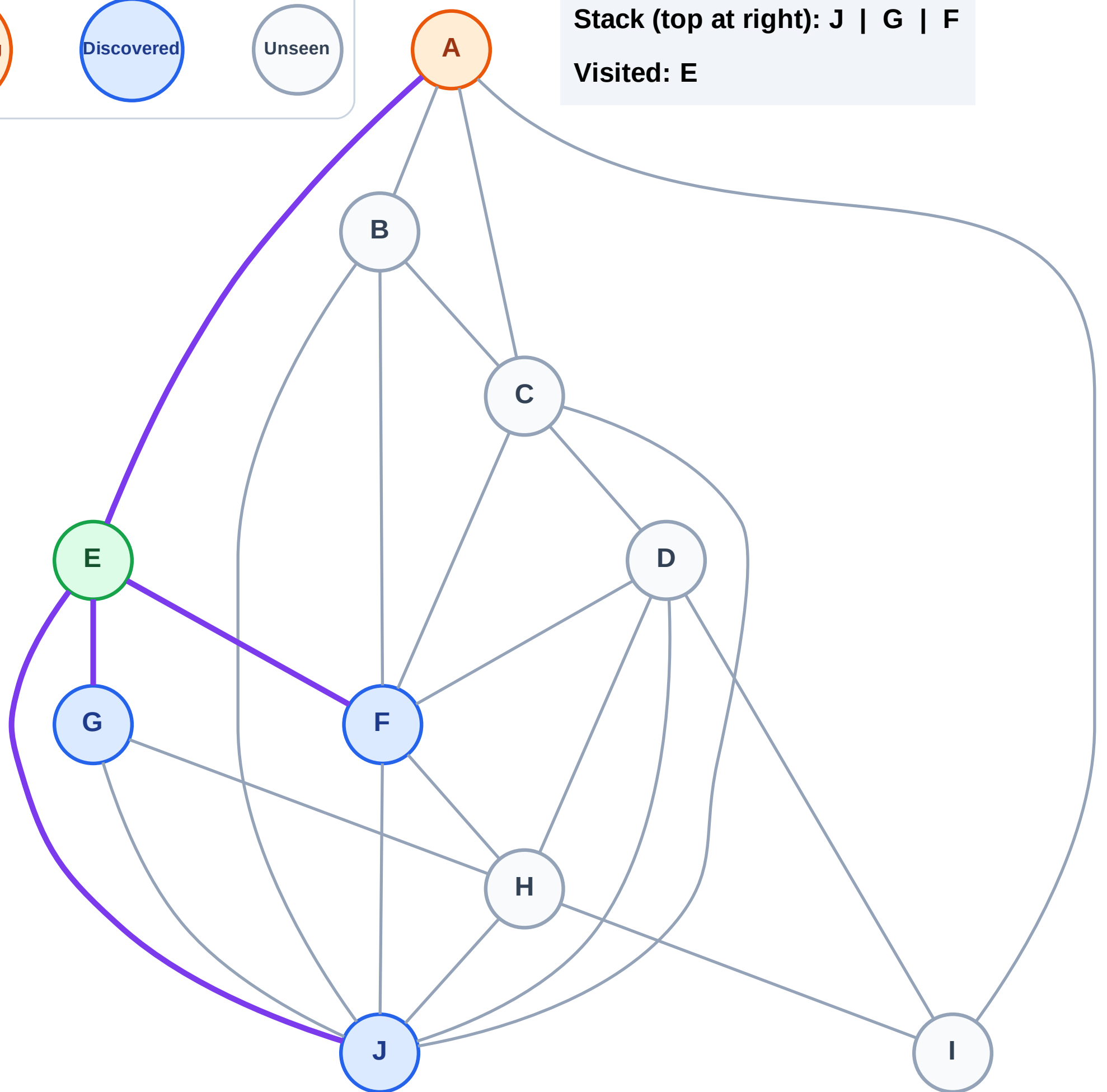
Step 4: Process node A

Legend



Stack (top at right): J | G | F

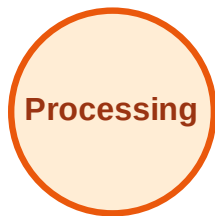
Visited: E



DFS Traversal

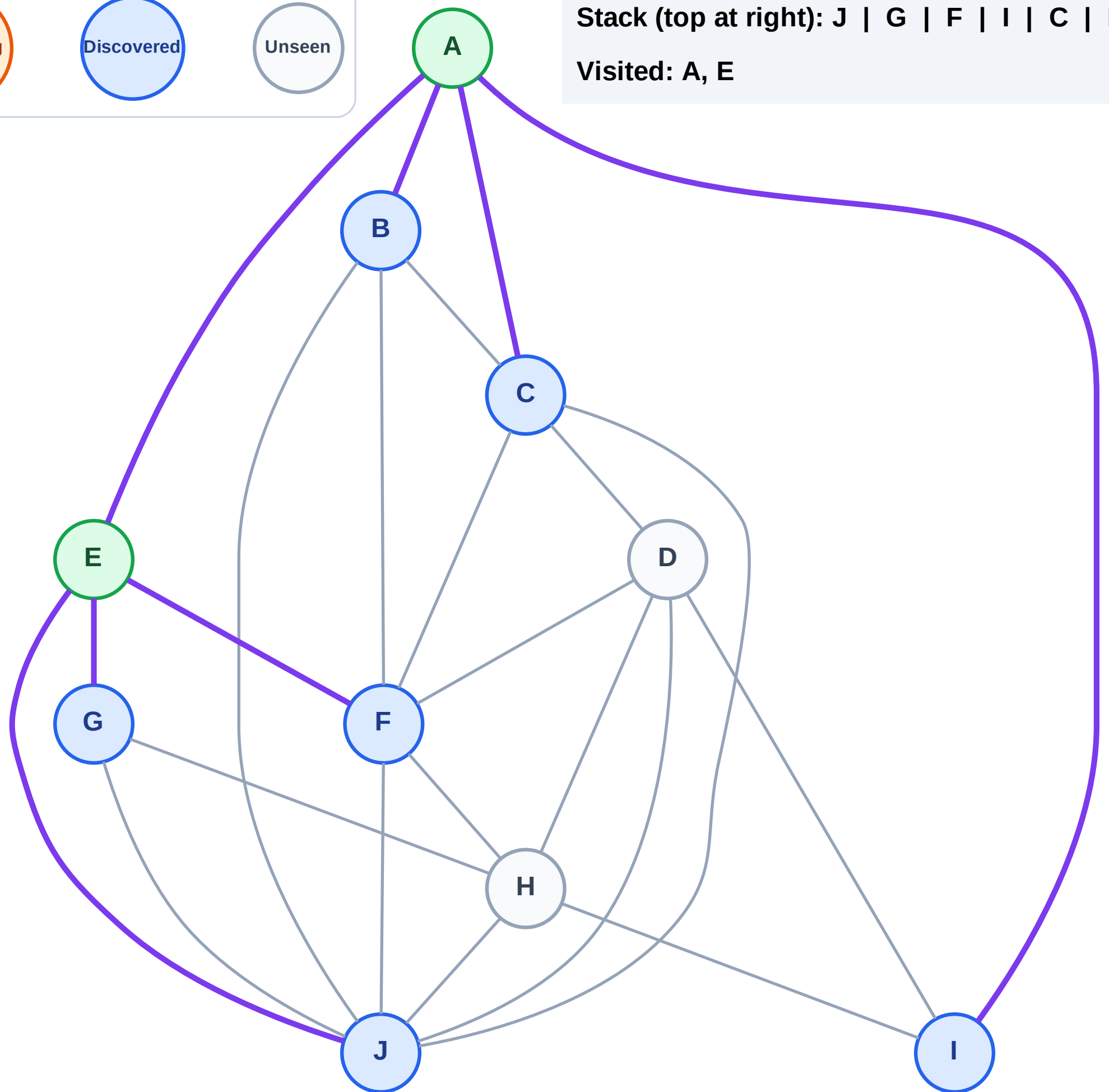
Step 5: Discover I, C, B from A

Legend



Stack (top at right): J | G | F | I | C | B

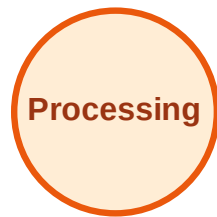
Visited: A, E



DFS Traversal

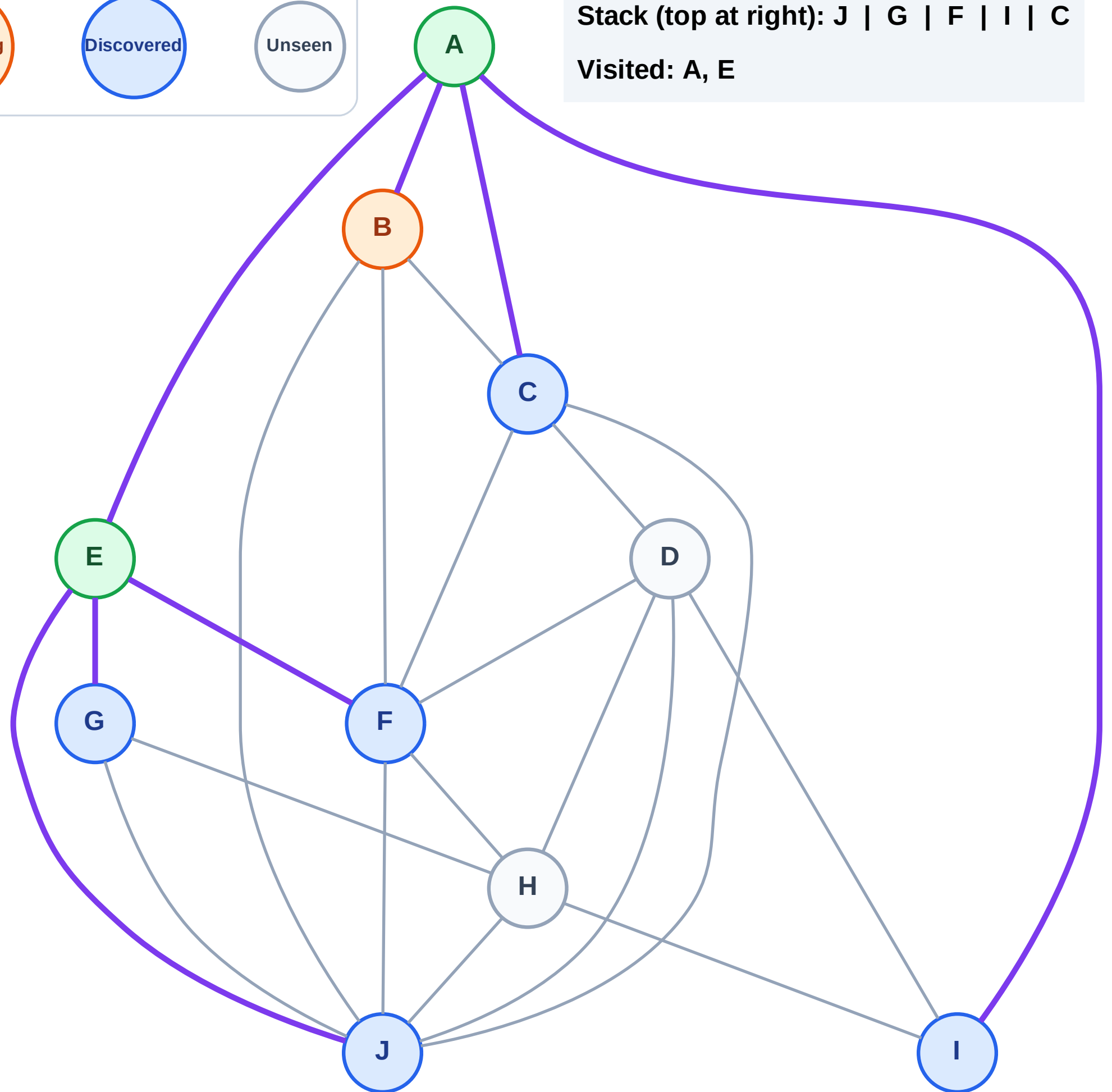
Step 6: Process node B

Legend



Stack (top at right): J | G | F | I | C

Visited: A, E

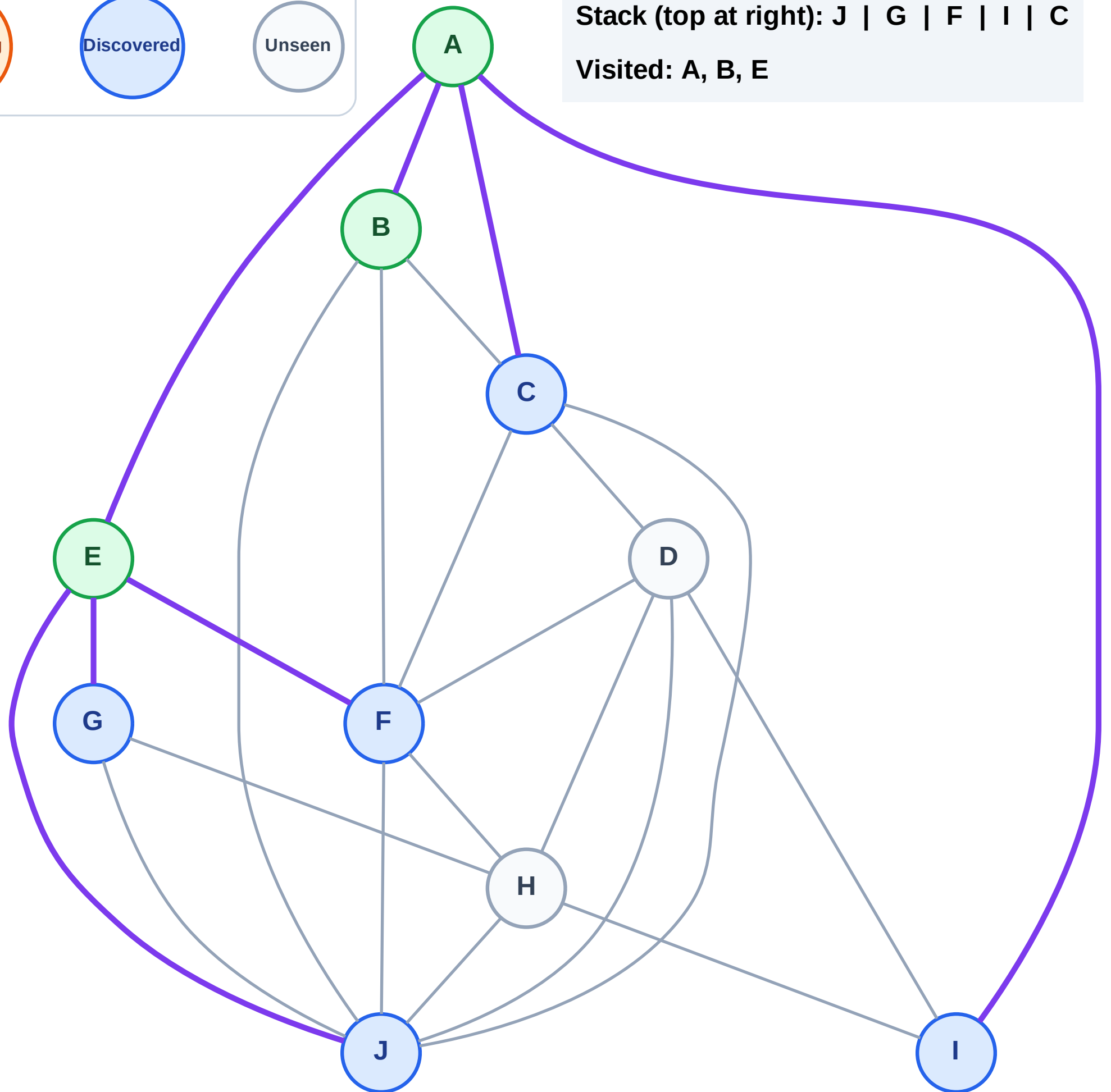


Step 7: No new neighbors from B

Legend

Unseen

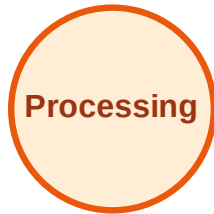
Visited: A, B, E



DFS Traversal

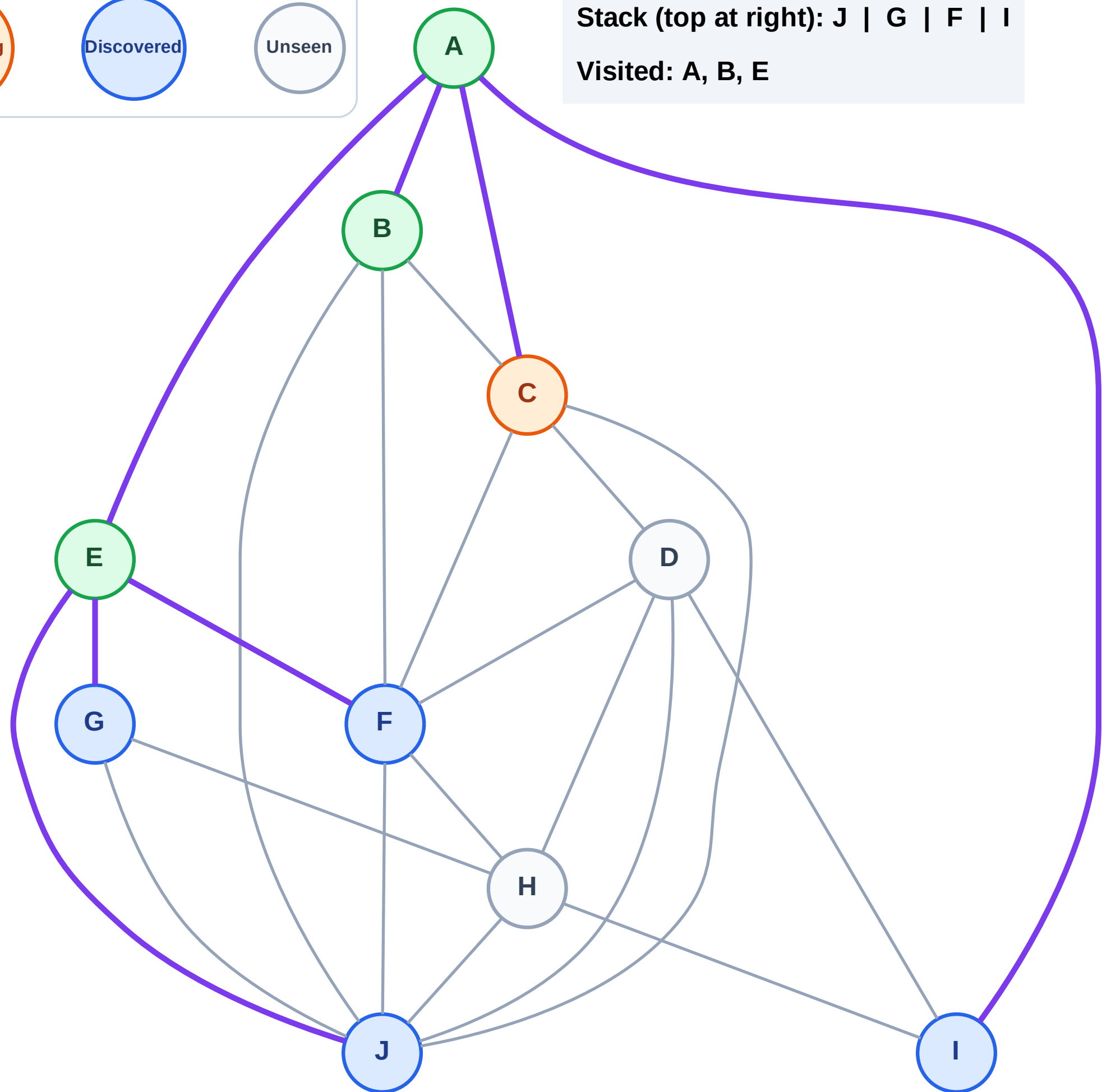
Step 8: Process node C

Legend



Stack (top at right): J | G | F | I

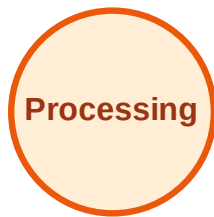
Visited: A, B, E



DFS Traversal

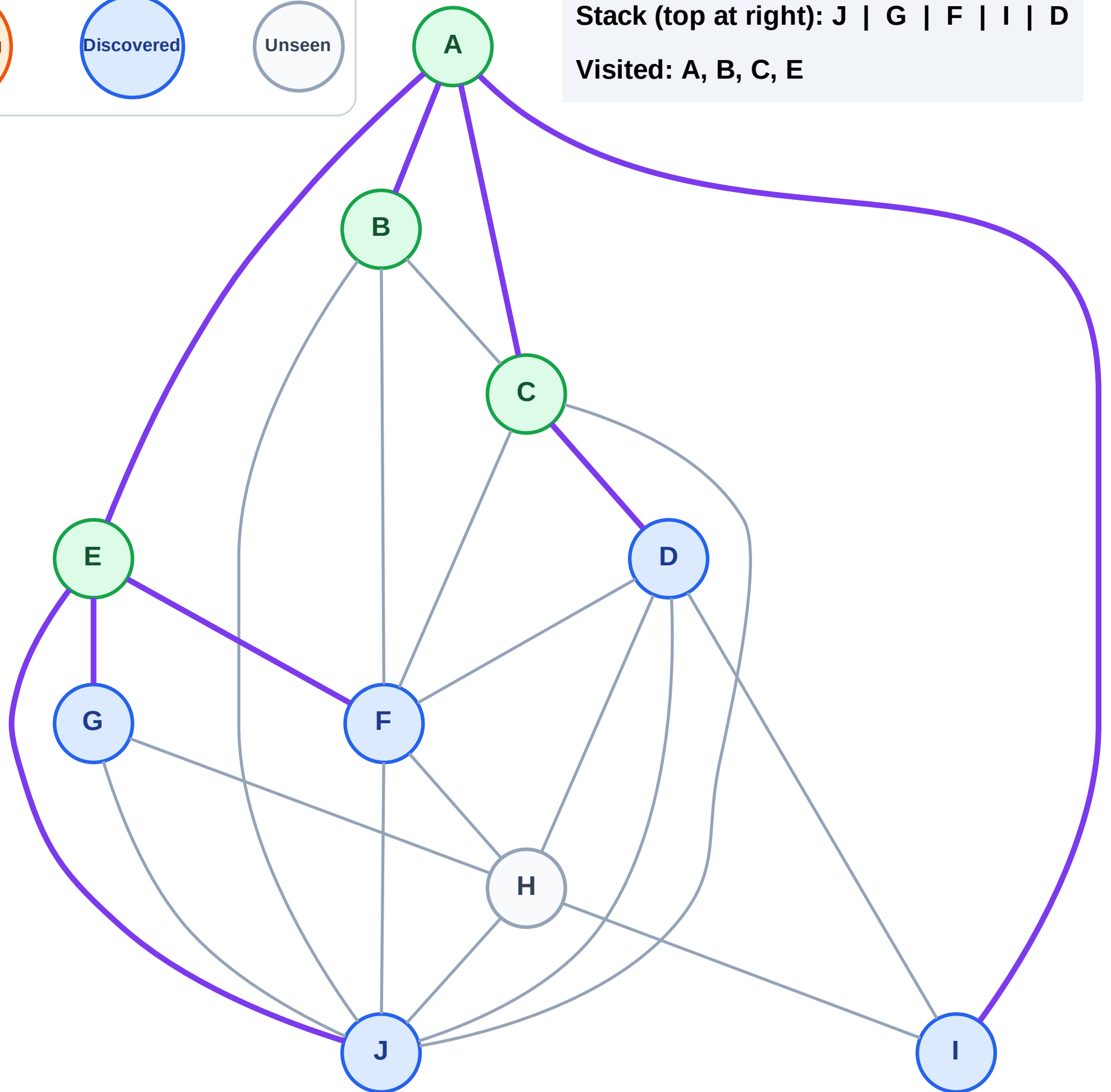
Step 9: Discover D from C

Legend



Stack (top at right): J | G | F | I | D

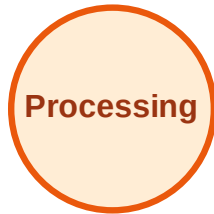
Visited: A, B, C, E



DFS Traversal

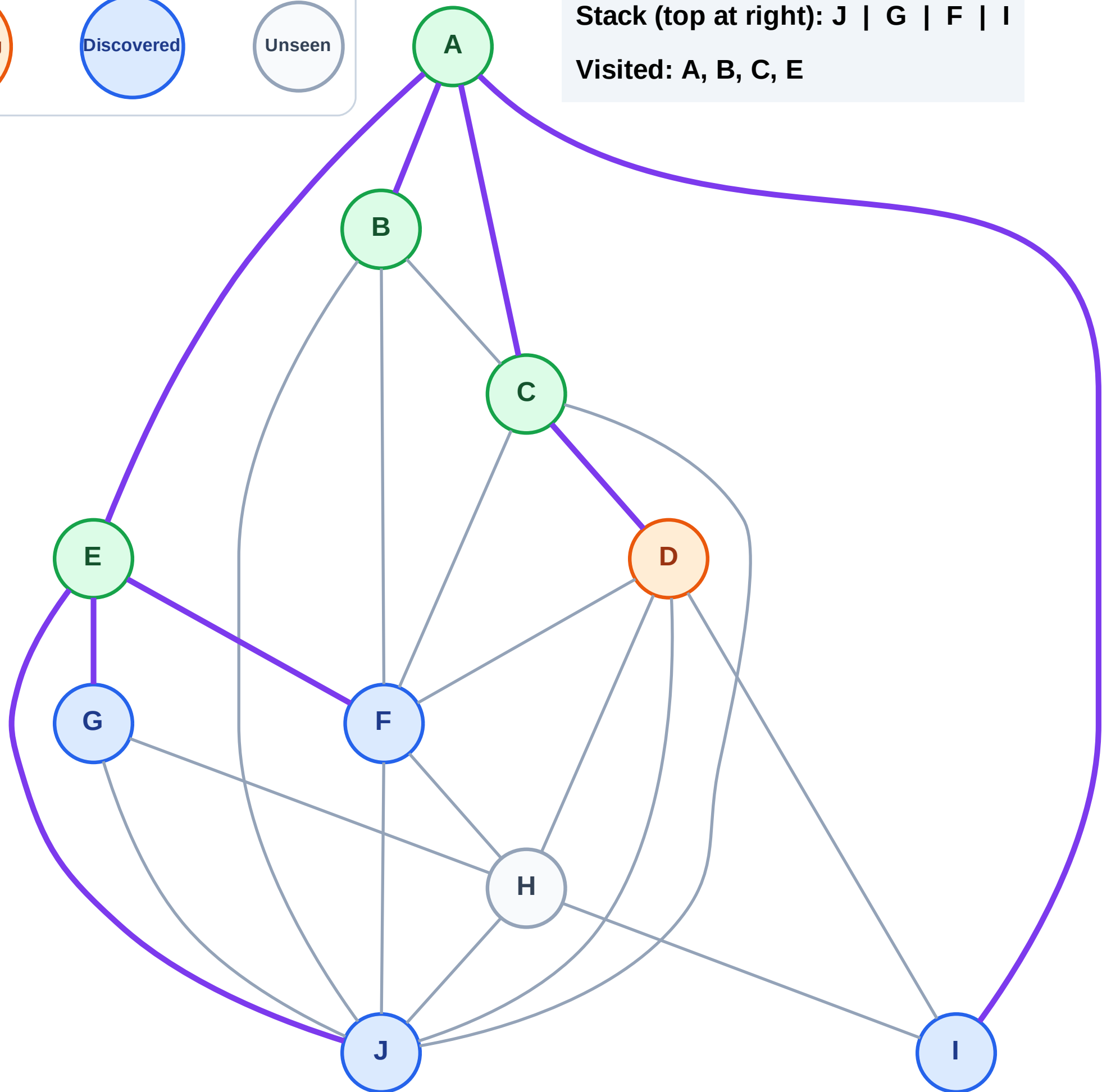
Step 10: Process node D

Legend



Stack (top at right): J | G | F | I

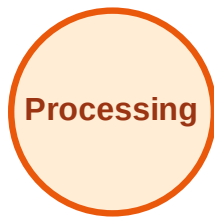
Visited: A, B, C, E



DFS Traversal

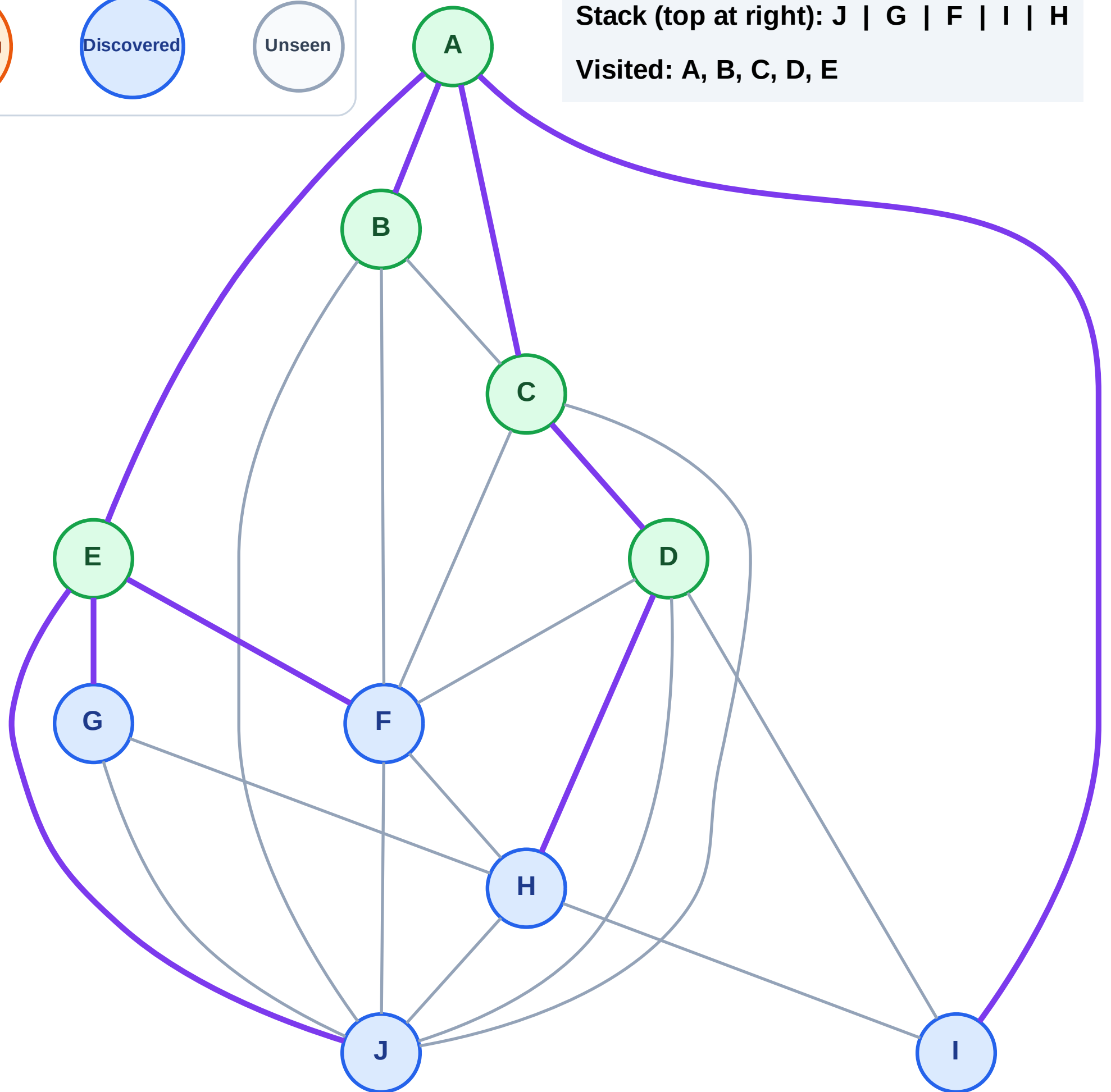
Step 11: Discover H from D

Legend



Stack (top at right): J | G | F | I | H

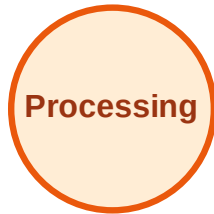
Visited: A, B, C, D, E



DFS Traversal

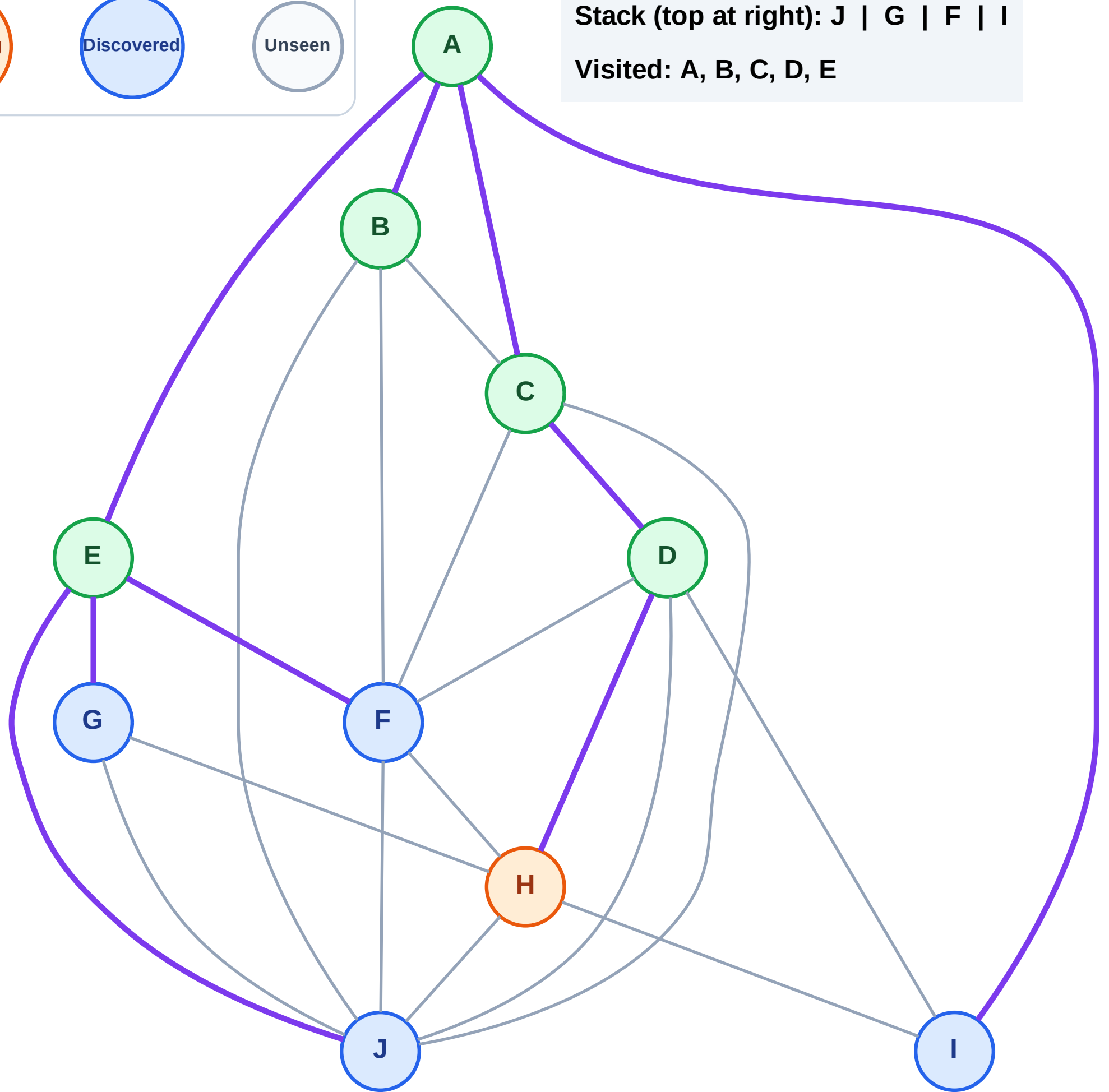
Step 12: Process node H

Legend



Stack (top at right): J | G | F | I

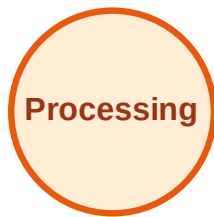
Visited: A, B, C, D, E



DFS Traversal

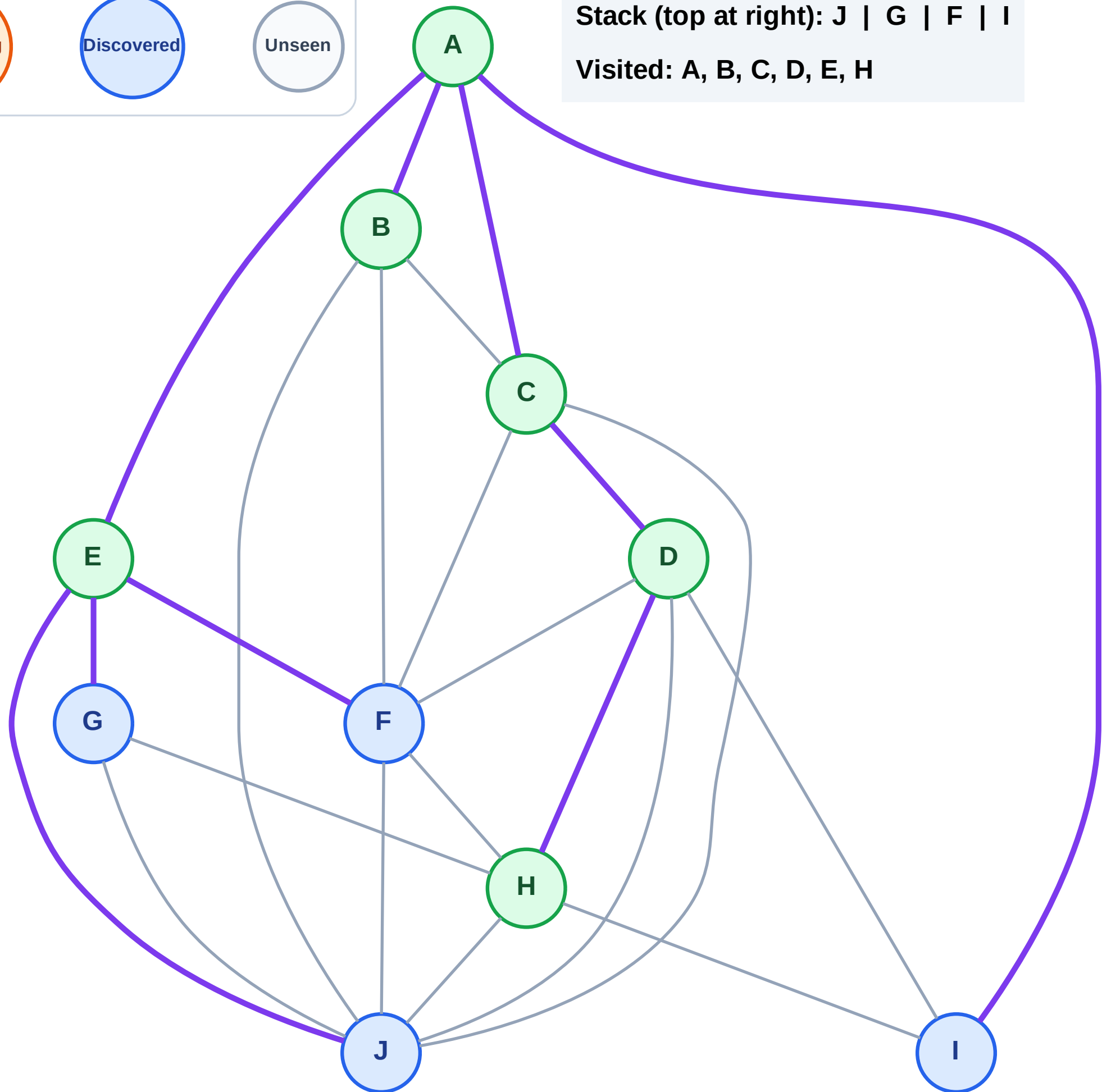
Step 13: No new neighbors from H

Legend



Stack (top at right): J | G | F | I

Visited: A, B, C, D, E, H



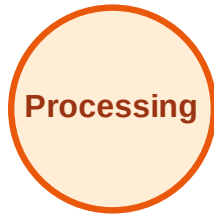
DFS Traversal

Step 14: Process node I

Legend



Complete



Processing



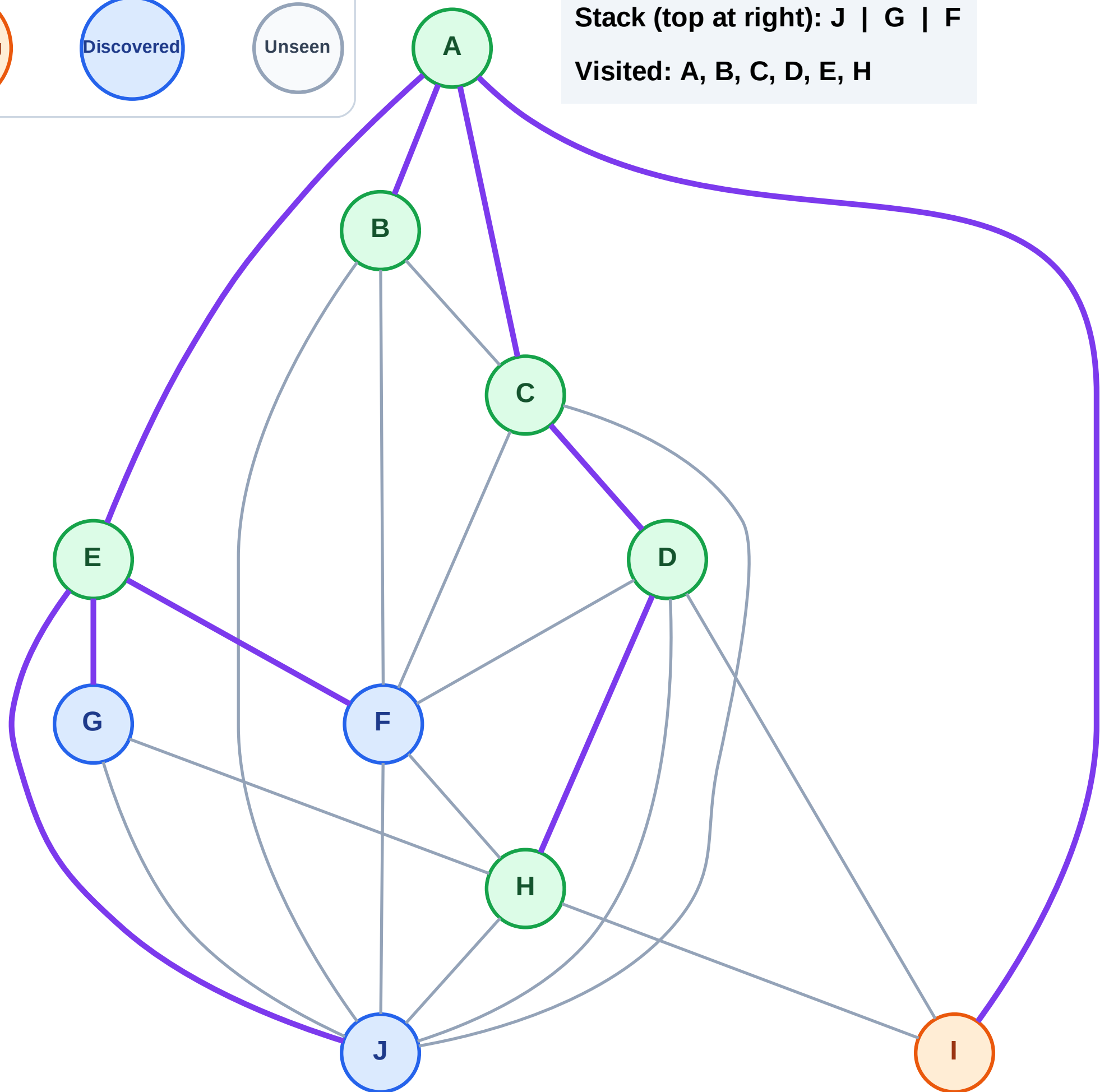
Discovered



Unseen

Stack (top at right): J | G | F

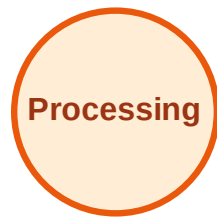
Visited: A, B, C, D, E, H



DFS Traversal

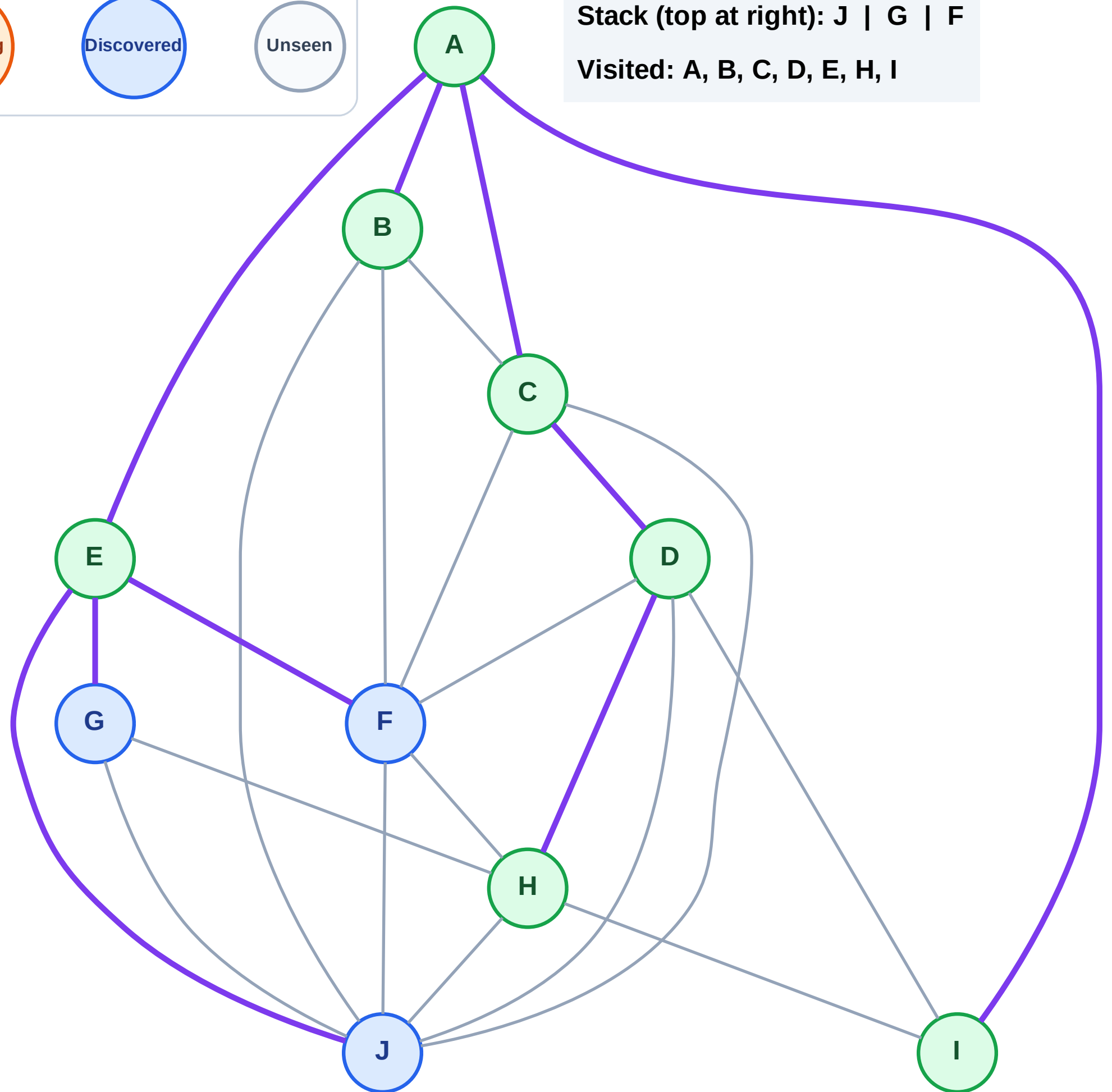
Step 15: No new neighbors from I

Legend



Stack (top at right): J | G | F

Visited: A, B, C, D, E, H, I



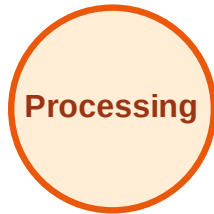
DFS Traversal

Step 16: Process node F

Legend



Complete



Processing



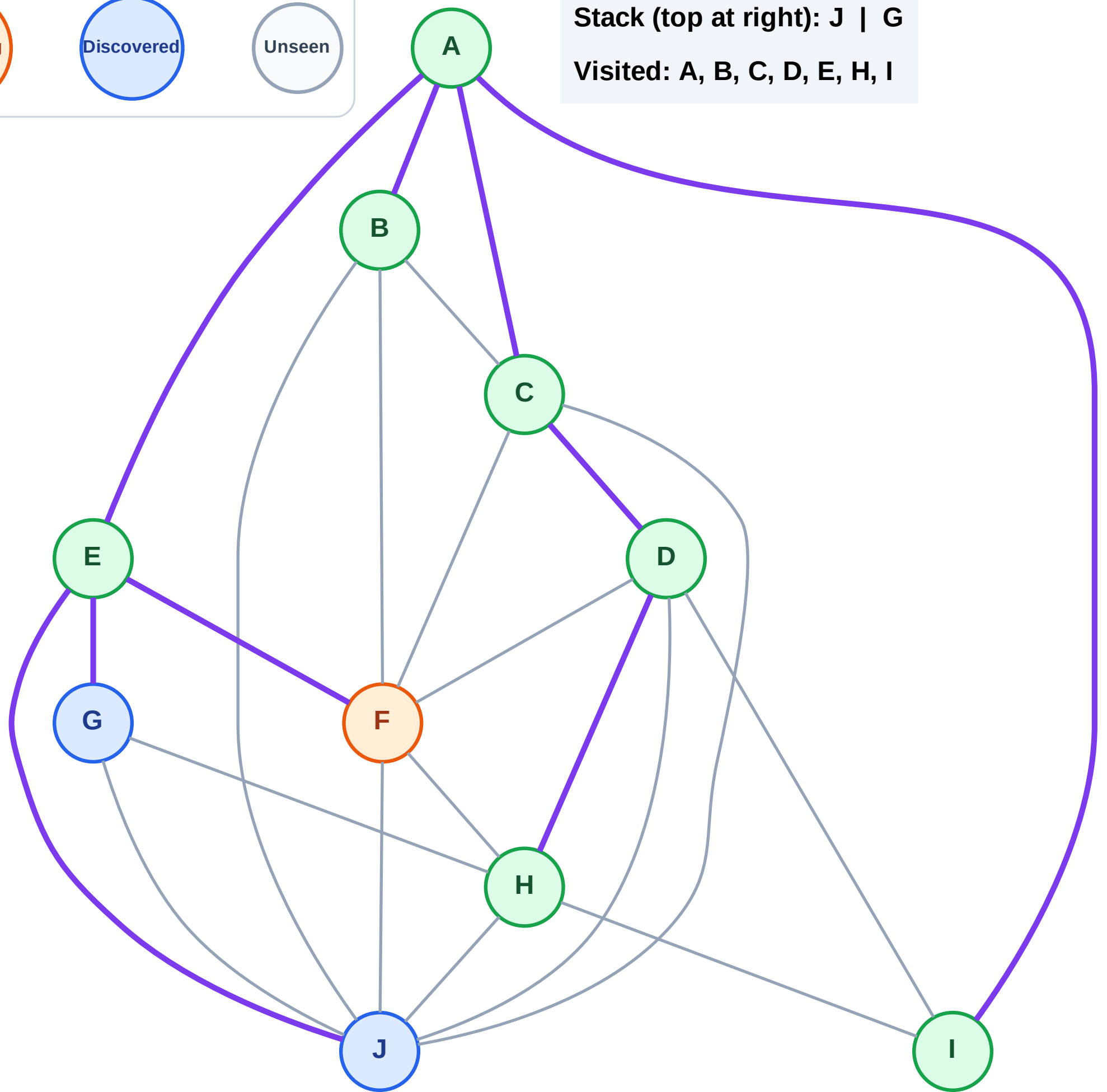
Discovered



Unseen

Stack (top at right): J | G

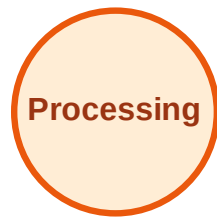
Visited: A, B, C, D, E, H, I



DFS Traversal

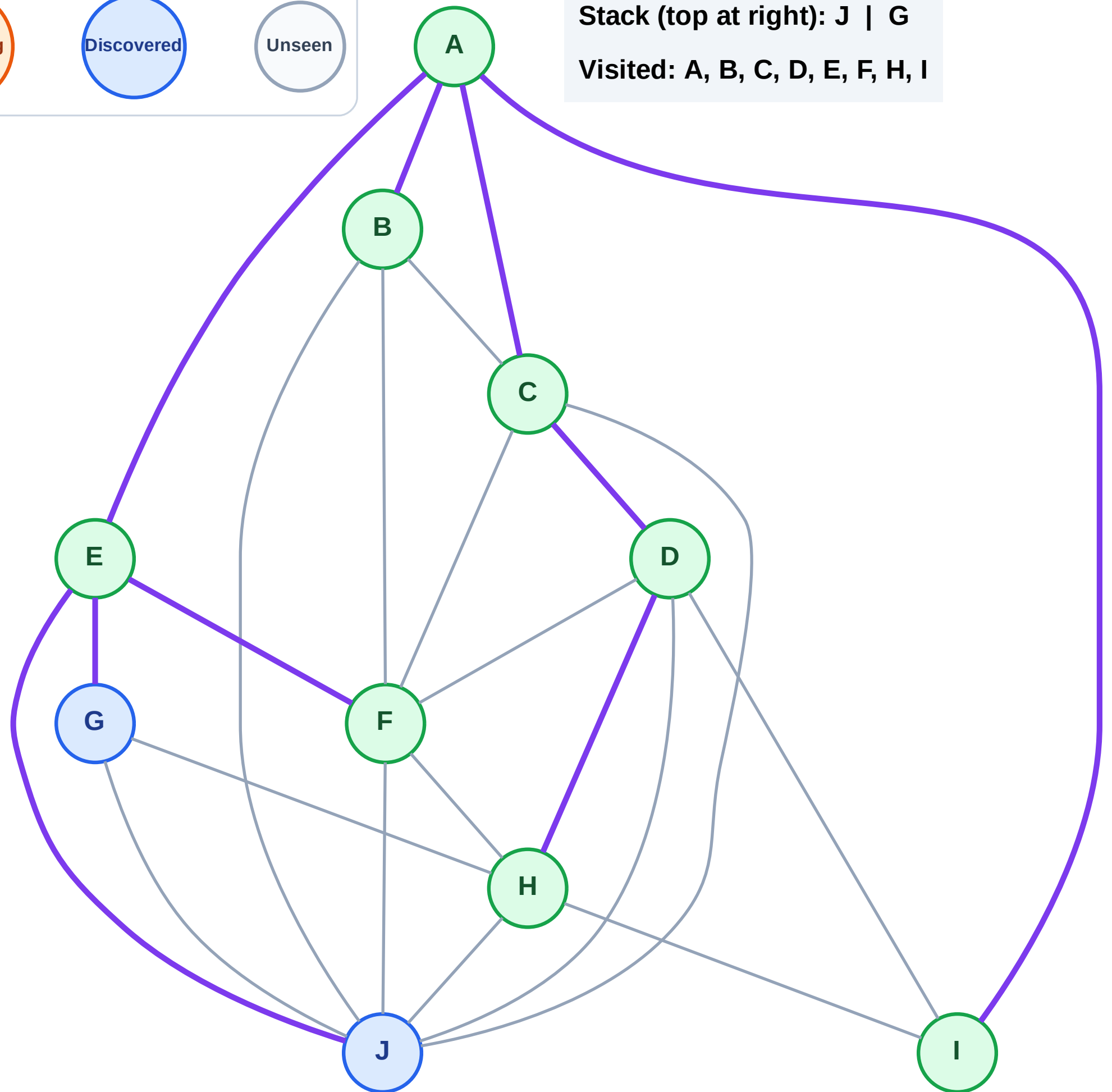
Step 17: No new neighbors from F

Legend



Stack (top at right): J | G

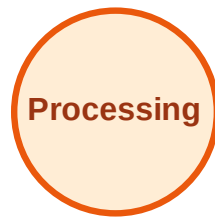
Visited: A, B, C, D, E, F, H, I



DFS Traversal

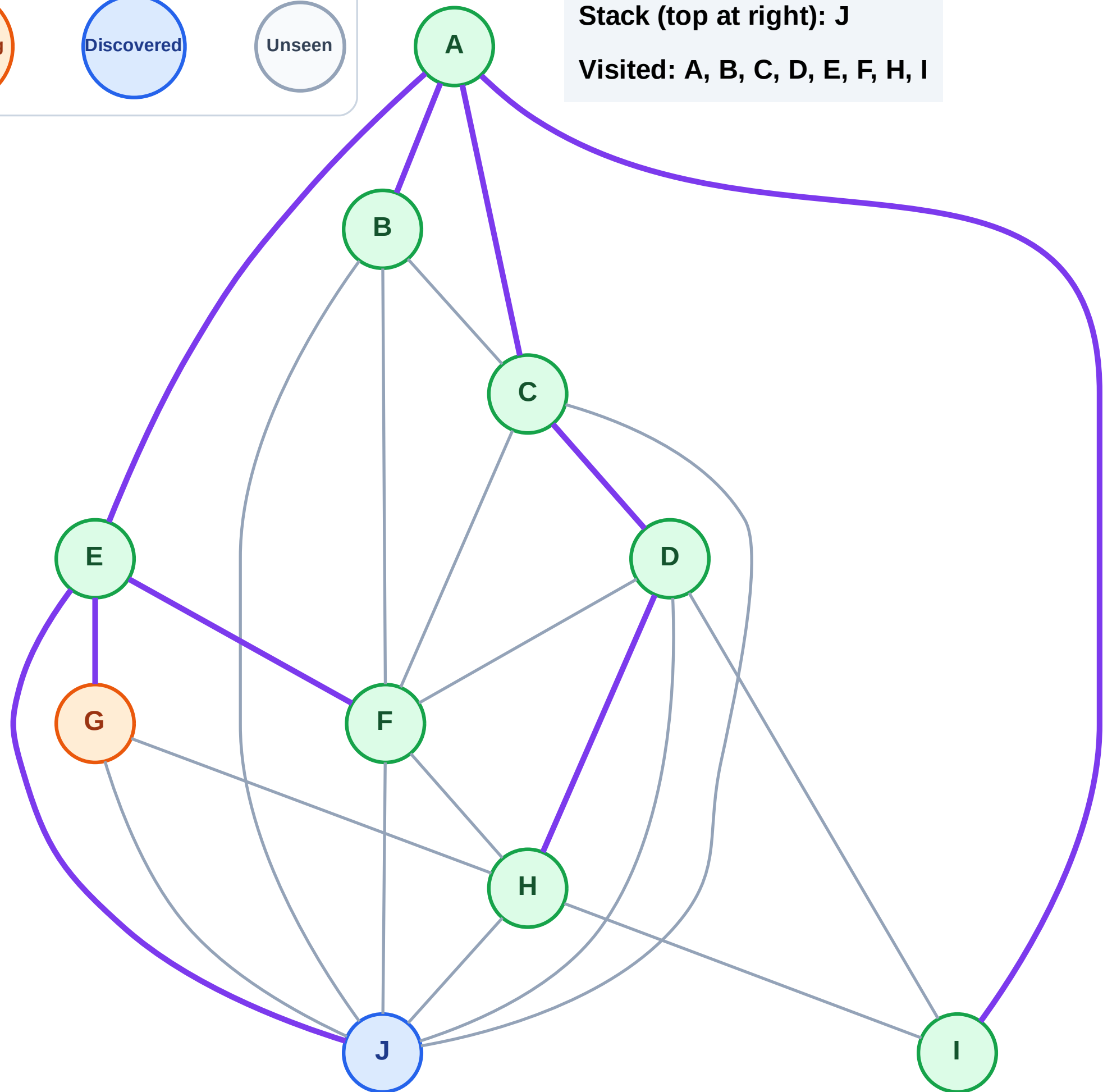
Step 18: Process node G

Legend



Stack (top at right): J

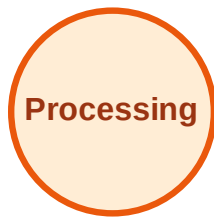
Visited: A, B, C, D, E, F, H, I



DFS Traversal

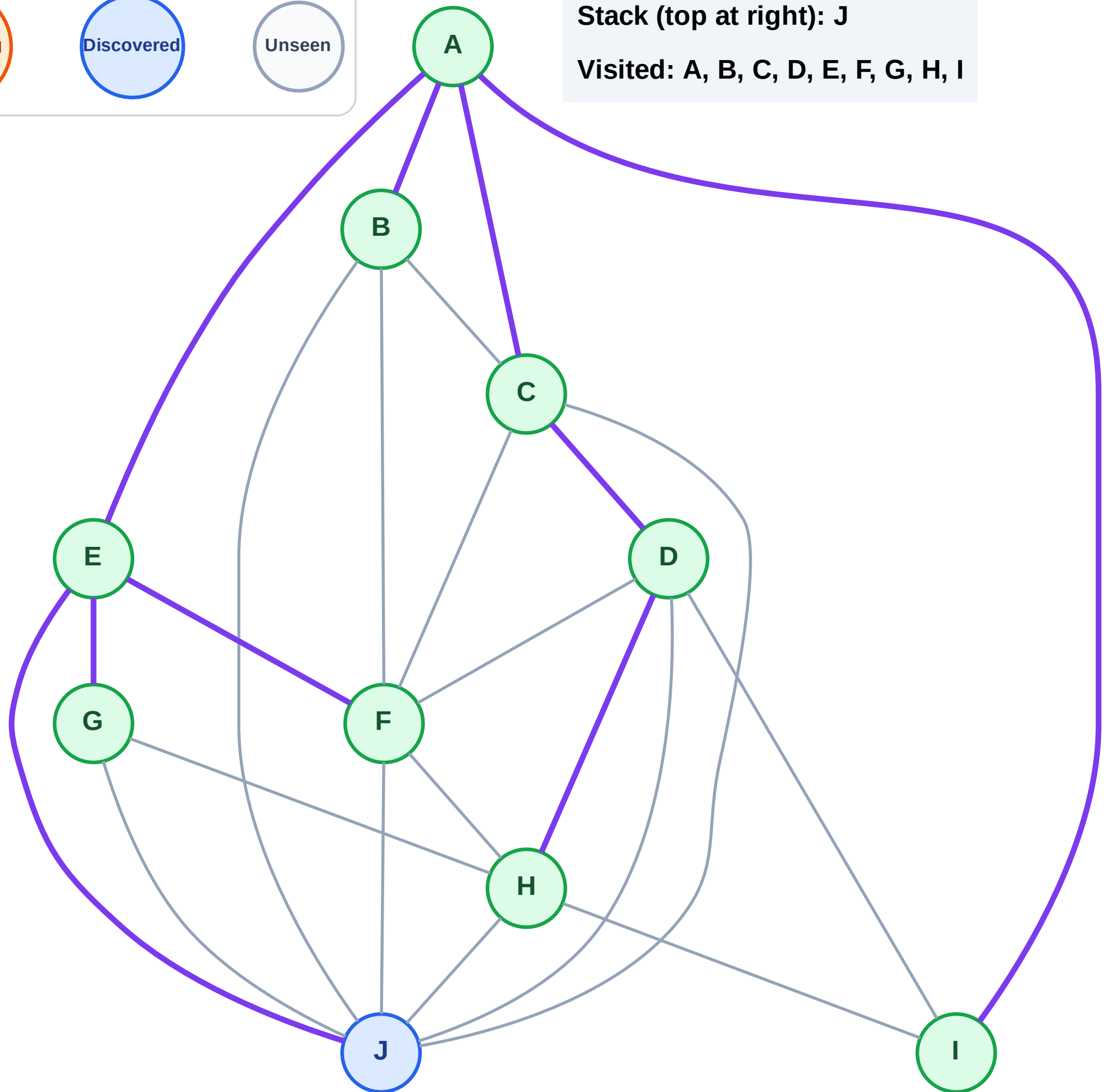
Step 19: No new neighbors from G

Legend



Stack (top at right): J

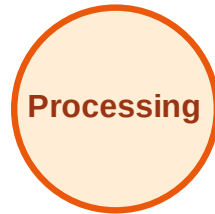
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

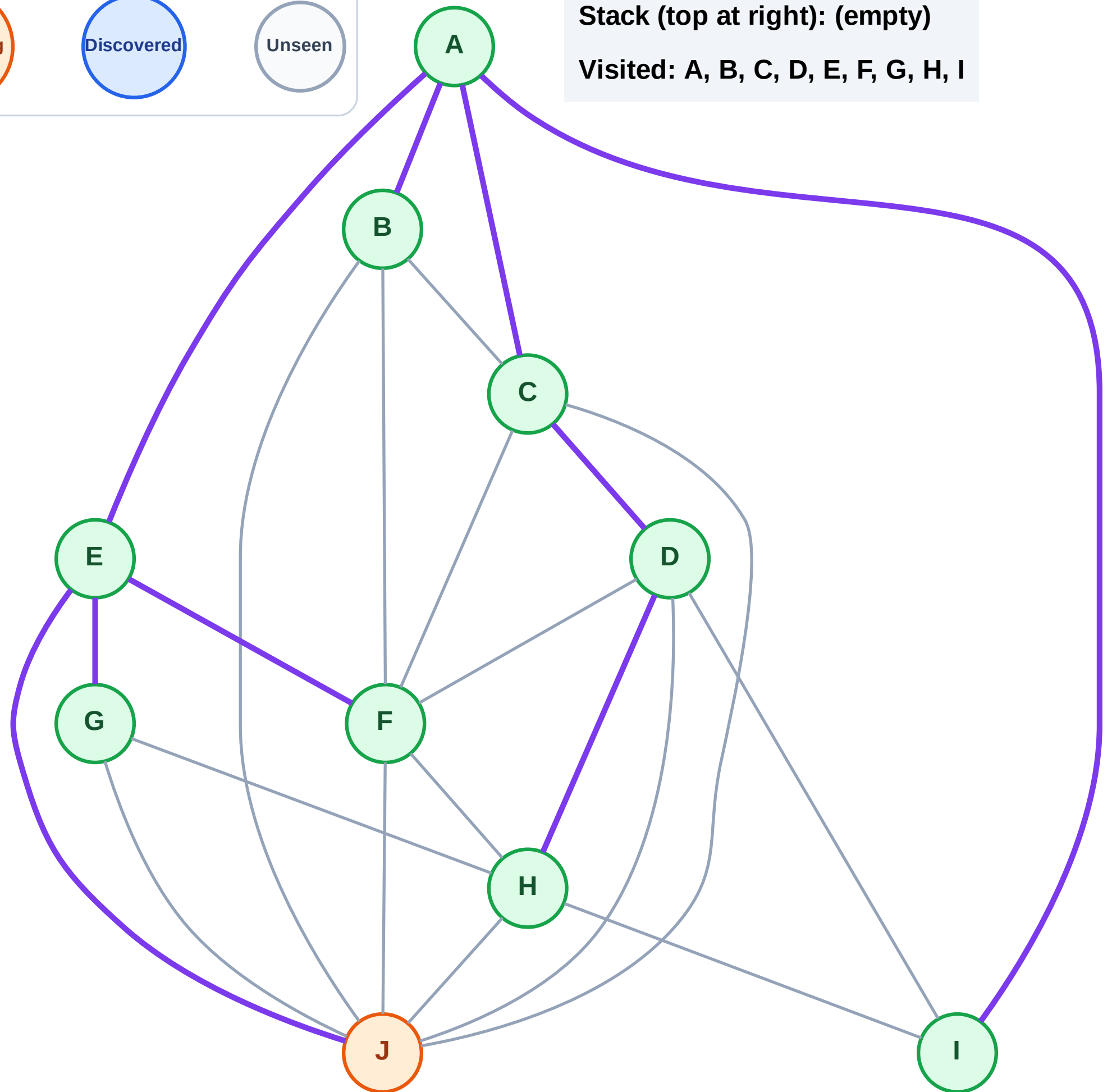
Step 20: Process node J

Legend



Stack (top at right): (empty)

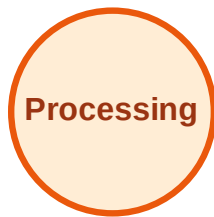
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

